International Journal of Accounting Information Systems 57 (2026) 100773

Contents lists available at ScienceDirect

International Journal of Accounting Information Systems

journal homepage: www.elsevier.com/locate/accinf

Smart accountability: leveraging AI to align ESG disclosure with practice

Kun Luo [a], Jiale Xue [a,*], Puspavathy Rassiah [b], Edwin KiaYang Lim [b]

[a] School of Economics and Management, Anhui Normal University, 241000 Wuhu, China
[b] Monash University Malaysia, Department of Accounting, School of Business, 47500 Bandar Sunway, Selangor Darul Ehsan, Malaysia

ARTICLE INFO

Keywords:
Artificial Intelligence
Learning-based AI
Logic-based AI
ESG Disclosure-Performance Alignment
Governance
Green Credit
China

ABSTRACT

This study examines whether artificial intelligence (AI) functions as a governance mechanism that reduces the misalignment between firms' environmental, social, and governance (ESG) disclosures and their underlying ESG performance. Using a sample of Chinese listed firms from 2011 to 2023, we find that AI adoption is associated with lower levels of ESG disclosure-performance misalignment, encompassing both the overstatement and understatement of ESG performance. Cross-sectional analyses further indicate that the association is more pronounced among large firms, non-state-owned enterprises, and firms operating in capital-intensive and technology-driven industries. Additional analysis shows that AI adoption is linked to improved access to green credit, with this relationship mediated by reductions in ESG disclosure-performance misalignment. Overall, this study contributes to the literature on AI and ESG by highlighting AI, particularly learning-based AI, can function as a governance-enabling technology that enhances the alignment between ESG disclosures and actual practices. Our study offers new insights into how different forms of AI (learning-based versus logic-based AI) influence ESG disclosure-performance alignment and capital market responses.

1. Introduction

Despite rising stakeholder expectations regarding firms' environmental, social, and governance (ESG) actions and disclosures (World Economic Forum, 2022; Gu et al., 2024; Krueger et al., 2024), substantial misalignment exists between what firms disclose and what they actually do. Some firms overstate or misrepresent their ESG performance, essentially "talking" more than "acting", which undermines trust and credibility. Others may, underreport their ESG efforts (i.e., "acting" more than "talking") due to reporting constraints or proprietary concerns, which hinders effective value communication (Delmas & Burbano, 2011). The ESG performance-disclosure misalignment[1] has significant economic and societal consequences, including erosion of stakeholder trust (Torelli et al., 2020), impaired corporate reputation (Wagner et al., 2009), and lower customer satisfaction (Ioannou et al., 2023). It is also associated

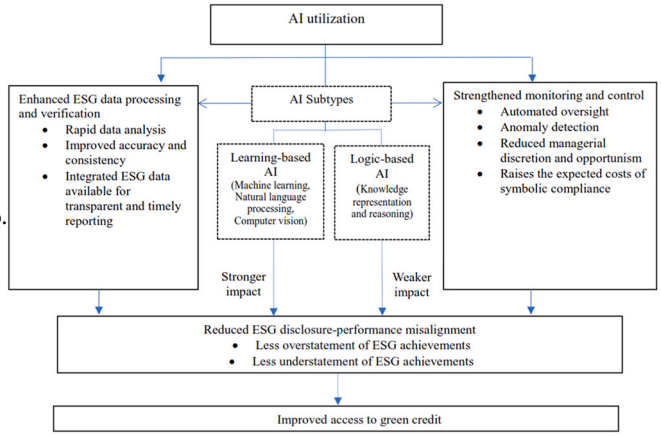
* Corresponding author.
E-mail addresses: luokunasd@ahnu.edu.cn (K. Luo), JialeX045@ahnu.edu.cn (J. Xue), Puspavathy.Rassiah@monash.edu (P. Rassiah), edwin.lim1@monash.edu (E.K. Lim).

[1] ESG disclosure-performance misalignment is exacerbated by disclosures that lack verifiable and quantifiable information, as well as by the largely voluntary nature of many ESG reporting frameworks, which enables selective disclosure. In response, governments and international standard-setting bodies, including the International Sustainability Standards Board, have introduced sustainability reporting standards aimed at improving disclosure quality.

https://doi.org/10.1016/j.accinf.2026.100773
Received 31 July 2025; Received in revised form 8 February 2026; Accepted 11 February 2026
Available online 17 February 2026
1467-0895/© 2026 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

with lower firm value (Du, 2015; Walker & Wan, 2012) and higher analyst forecast errors, which, in turn, restrict access to external financing (García-Sánchez et al., 2021). These concerns give rise to a fundamental question: how can the gap between ESG "talking" and "acting" be meaningfully narrowed? We address this question by investigating whether artificial intelligence (AI),[2] and its distinct subtypes, can serve as a governance-enhancing force that strengthens the alignment between firms' ESG disclosures and their underlying practices.

Governments worldwide have introduced strategic policies to promote the development and adoption of AI across the economy.[3] Consistent with this trend, interest in AI has grown steadily within both the accounting profession and academia (Sutton et al., 2016; Stratopoulos & Wang, 2025).[4] We posit a negative relationship between AI utilization and ESG disclosure-performance misalignment, driven by two interrelated mechanisms. These effects are expected to vary by AI type and to manifest in tangible economic outcomes, particularly firms' access to green credit, as illustrated in the conceptual map.



Conceptual Map: AI Utilization, ESG Performance-Disclosure misalignment and green credit

First, AI is expected to reduce ESG disclosure-performance misalignment by enhancing data processing and verification. By enabling the rapid analysis of large and complex datasets, AI improves firms' information-processing and decision-making capabilities, leading to greater data accuracy, completeness, and internal consistency (Commerford et al., 2022; Holmes & Douglass, 2022; Babina et al., 2024). Improved generation, integration, and verification of ESG-related data enhance data quality and transparency (Agrawal et al., 2019; Chen et al., 2024). As a result, AI limits ESG overstatement by constraining firms' ability to exaggerate ESG performance without substantiation and reduces ESG understatement by improving firms' capacity to accurately capture and disclose substantive ESG actions.

Second, AI strengthens monitoring and constrains managerial discretion. ESG disclosure-performance misalignment is commonly explained by agency, stakeholder, and legitimacy theories as arising from information asymmetry, heterogeneous stakeholder pressures, and symbolic compliance in response to legitimacy demands (Delmas & Burbano, 2011; Walker & Wan, 2012). Managers may exploit discretion to selectively disclose ESG information by overstating favorable outcomes or withholding negative information. AI mitigates these frictions by reducing reliance on human judgment and enhancing the credibility of ESG information (N. Li et al., 2024; S. Li et al., 2024). Through automated monitoring, traceability, and anomaly detection (Kellogg et al., 2020), AI increases the visibility of ESG-related activities and raises the expected costs of symbolic compliance. As a result, AI reduces opportunities for ESG overstatement. In parallel, AI mitigates ESG understatement by improving internal coordination and systematically capturing genuine ESG efforts, thereby aligning ESG disclosures more closely with underlying practices.

ESG disclosure-performance misalignment signals to markets and regulators that a firm's sustainability commitments lack substance. This can erode credibility, heighten perceived ESG and financing risk, and constrain access to green finance. In this context, AI can function as a governance mechanism that reduces information and ESG risks and enhances the credibility of ESG practices. By mitigating ESG disclosure-performance misalignment, we posit that AI ultimately improves firms' access to green credit.

Building on these mechanisms, we hypothesize that learning-based AI is more effective in reducing ESG disclosure-performance misalignment than logic-based AI. ESG misalignment often manifests in subtle, evolving, and context-dependent ways that extend beyond explicit rule violations. By learning probabilistic patterns from large volumes of unstructured text, operational data, and

[2] The Organization for Economic Co-operation and Development (OECD) defines AI as systems that generate predictions, recommendations, or decisions to achieve human-defined objectives by emulating core cognitive functions such as learning and reasoning under uncertainty (Simon, 1995).

[3] For example, China launched the *Guidelines for the Construction of National Artificial Intelligence Innovation and* Development Pilot Zones in 2019 to establish experimental hubs that integrate AI into real-world business applications. Similarly, the United States introduced the National Artificial Intelligence Research and Development Strategic Plan in 2016, prioritizing national security and economic competitiveness through AI-driven innovation (Hine & Floridi, 2024).

[4] In the sustainability domain, prior studies show that AI adoption fosters green innovation (Tian et al., 2023), alleviates green financing constraints (Zhong & Song, 2025), reduces carbon emissions (Chen et al., 2022), and enhances overall ESG performance (S. Li et al., 2024; Chen & Zhang, 2025; J. Li et al., 2025).

external information (Townsend et al., 2025), learning-based AI, such as machine learning, natural language processing, and computer vision, can detect inconsistencies, exaggeration, and selective omission in sustainability narratives while also uncovering substantive but under-disclosed ESG actions (Agrawal et al., 2018; Sari et al., 2025). In contrast, logic-based AI, which relies on predefined rules and symbolic reasoning, is well suited for enforcing formal disclosure compliance but is less adaptable to evolving ESG practices due to its dependence on codified knowledge and fixed rules (Boden, 2014; Chollet, 2019).

To implement our analysis, we conduct textual analysis of annual reports to measure firms' AI usage, and define ESG disclosure-performance misalignment as the difference between standardized ESG disclosure quality scores from Bloomberg and standardized ESG performance scores from LSEG (formerly Refinitiv). Using a sample of 12,850 firm-year observations from Chinese A-share listed firms over the period 2011–2023, we find that greater AI utilization is associated with significantly lower ESG disclosure-performance misalignment. This relation holds for both ESG overstatement and ESG understatement. Our findings are robust across a set of endogeneity and sensitivity checks.

Importantly, AI utilization retains strong explanatory power even after controlling for firms' broader digital transformation efforts. The estimated coefficient on AI utilization is approximately 5.1 times larger than that of digital transformation, indicating that AI provides incremental informational and governance benefits beyond general digitalization. Whereas digital transformation primarily reflects the adoption of digital tools and infrastructure to improve operational efficiency, AI represents an advanced governance capability that learns from data, generates predictive insights, and can autonomously influence organizational decisions. This evidence suggests that AI plays a distinct role in reducing ESG disclosure-performance misalignment, extending prior findings reported by Chen et al. (2024).

Further analyses show that the reduction in ESG disclosure-performance misalignment associated with AI is driven primarily by learning-based AI rather than logic-based AI. The mitigating effect of AI is more pronounced among large firms, non-state-owned enterprises, and firms in capital-intensive and technology-intensive industries, where complementary infrastructure, expertise, and organizational capacity support more effective AI implementation. Finally, AI utilization is positively associated with firms' access to green credit, mediated by reduced ESG disclosure-performance misalignment.

Our study makes several academic contributions. First, we advance research on AI and ESG by conceptualizing AI as a technology-enabled internal mechanism that mitigates ESG disclosure-performance misalignment. Prior studies primarily emphasize traditional institutional mechanisms, including disclosure mandates (Aboud et al., 2024), external monitoring by analysts and institutional investors (Wang et al., 2024; Eliwa & Elmaghrabi, 2025), and board-level governance reforms (Gull et al., 2023a; Gull et al., 2023b), while paying limited attention to the role of AI in addressing ESG disclosure-performance misalignment. By shifting the focus from external enforcement to process-oriented internal governance, our study fills this important gap.

Our study differs meaningfully from Chen et al. (2024), who examine the effect of broad digital transformation on ESG disclosure-performance misalignment, without isolating AI or distinguishing among AI types. In addition, prior studies focusing on the relationship between AI and ESG performance scores or ratings (S. Li et al., 2024; J. Li et al., 2025; Chen & Zhang, 2025; Lin & Zhu, 2025; Zhou et al., 2025) do not examine ESG disclosure-performance misalignment, nor do they differentiate between AI subtypes. We extend this literature by explicitly positioning AI as a governance mechanism that reduces ESG disclosure-performance misalignment and by distinguishing between learning-based and logic-based AI, thereby offering new insights into how different AI subtypes influence ESG outcomes. In addition, by distinguishing between ESG overstatement and ESG understatement, we provide a more nuanced understanding of how AI simultaneously constrains symbolic reporting, encourages substantive ESG actions, and induces commensurate ESG disclosures that create value.

Second, we document the economic consequences of AI-enabled governance by showing that AI utilization, through reduced ESG disclosure-performance misalignment, is associated with greater access to green finance. This evidence links AI-enabled governance not only to improved reporting outcomes but also to tangible capital market benefits. Finally, we reveal heterogeneous effects of AI adoption across firm characteristics, including firm size, industry, and technological intensity. Whereas prior research emphasizes governance structures and managerial intent (Chen et al., 2024; De Villiers & Marques, 2016), our findings underscore the conditional role of organizational capacity and technological readiness in shaping the effectiveness of AI-enabled governance in reducing ESG disclosure-performance misalignment across diverse settings.

In summary, prior research shows that digital transformation is broadly associated with reduced ESG disclosure-performance misalignment (Chen et al., 2024), while AI adoption is linked to higher ESG ratings and performance (S. Li et al., 2024; J. Li et al., 2025; Chen & Zhang, 2025; Lin & Zhu, 2025; Zhou et al., 2025). However, this literature either subsumes AI within an undifferentiated digital transformation construct or focuses on ESG performance outcomes in isolation, without examining how AI influences the alignment between ESG actions and disclosures. Addressing this gap, we conceptualize AI as a distinct governance technology, differentiate learning-based and logic-based AI, and develop a causal framework linking AI utilization to ESG disclosure-performance misalignment and its economic consequences, particularly firms' access to green credit. Our findings show that AI, especially learning-based AI, reduces ESG disclosure-performance misalignment, thereby enhancing firms' credibility with capital providers and improving access to green financing.

2. Literature Review and Hypotheses Development

2.1. ESG Disclosure-Performance Misalignment

The growing emphasis on ESG disclosure has intensified concerns about the misalignment between firms' ESG disclosures ("talk") and their underlying ESG practices ("action") (Delmas & Burbano, 2011). Agency theory (Jensen & Meckling, 1976) helps explain the

persistence of ESG disclosure-performance misalignment, as managers possess superior private information and may selectively disclose or manipulate ESG information when incentives diverge from those of principals. When monitoring is imperfect, managerial discretion allows firms to portray favorable ESG profiles while obscuring adverse outcomes (Kim & Lyon, 2015). The high discretion and low verifiability of ESG disclosures further exacerbate information asymmetry and opportunistic reporting (Christensen et al., 2021).

Stakeholder theory complements this view by highlighting the complexity of firms' relationships with multiple stakeholder groups that hold heterogeneous and evolving expectations (Freeman, 1984; Mitchell et al., 1997). On the one hand, firms are frequently criticized for relying on symbolic ESG communication rather than substantive sustainability actions to shape stakeholder perceptions (Walker & Wan, 2012). On the other hand, there are also firms deliberately downplaying or withholding information about their ESG achievements to mitigate heightened stakeholder expectations and scrutiny (Carlos & Lewis, 2018).

From a legitimacy theory perspective, ESG disclosures serve as symbolic mechanisms through which firms seek to demonstrate alignment with prevailing social norms and societal expectations (Deegan, 2002). As legitimacy pressures intensify, firms may increasingly rely on disclosure-based strategies to signal conformity, without undertaking substantive operational or strategic changes, thereby reinforcing ESG disclosure-performance misalignment (Delmas & Burbano, 2011).

Prior research suggests that external monitoring by institutional investors, regulators, and the media can, paradoxically, intensify firms' incentives to overstate sustainability performance, thereby reinforcing ESG disclosure-performance misalignment (Topal et al., 2020; Tang et al., 2023). Institutional investors, in particular, may either constrain or exacerbate ESG disclosure-performance misalignment depending on their monitoring intensity, engagement strategies, and investment horizons (Eliwa & Elmaghrabi, 2025). Internally, governance structures characterised by lower CEO power concentration, greater board independence, and higher gender diversity are generally associated with enhanced transparency and reduced ESG disclosure-performance misalignment (Gull et al., 2023a; Gull et al., 2023b).

Despite the presence of conventional governance mechanisms such as external monitoring, board oversight, and disclosure mandates, their effectiveness in curbing symbolic ESG behavior remains limited. These mechanisms struggle to keep pace with the volume and complexity of ESG data, particularly across global supply chains and multiple regulatory regimes, and rely heavily on human judgment and *ex post* verification (Aboud et al., 2024). Consequently, monitoring gaps persist, allowing ESG disclosure-performance misalignment to endure. This highlights the need for more systematic, scalable, and near real-time AI-enabled governance mechanisms.

2.2. *Artificial Intelligence (AI)*

Since McCulloch and Pitts' (1943) foundational work on neural computation, AI has evolved from rule-based symbolic systems to data-driven, adaptive learning technologies. According to Boritz and Stratopoulos (2023), AI has advanced through three distinct phases: early symbolic approaches from the late 1950 s to the 1970 s, expert systems in the 1980 s, and a data-driven era since the late 1990 s enabled by big data and cloud computing. The current surge in AI adoption, driven by advances in machine learning and deep neural networks, has elevated AI to a leading frontier technology and expanded its role beyond efficiency gains to strategic decision-making and organizational governance (Kaplan & Haenlein, 2019).

Empirically, AI adoption is associated with reduced information frictions and enhanced productivity, resulting in positive outcomes such as increased international trade and innovative capacity (Brynjolfsson et al., 2019; Babina et al., 2024). Beyond efficiency gains, AI increasingly functions as a strategic governance resource by automating routine tasks, reducing human discretion, improving process reliability, and enabling continuous, data-driven monitoring (Kolbjørnsrud et al., 2016; Kellogg et al., 2020; Holmes & Douglass, 2022). Through enhanced anomaly detection, superior prediction, and improved judgment consistency, recent studies highlight AI's relevance in improving accounting estimates, predicting misstatements and facilitating audit effectiveness (Ding et al., 2020; Christ et al., 2021; Commerford et al., 2022; Fedyk et al., 2022; Zhang et al., 2022; Han et al., 2023; Zhang Parker et al., 2025), as well as sustainability related outcomes, including improved ESG monitoring, performance and assurance (N. Li et al., 2024; S. Li et al., 2024; J. Li et al., 2025; Chen & Zhang, 2025; Lin & Zhu, 2025; Zhou et al., 2025).[5]

While digital transformation broadly captures firms' adoption of digital infrastructure and applications (e.g., cloud computing, digital platforms, blockchain, and smart manufacturing systems), AI represents a distinct, higher-order governance technology characterized by learning, prediction, and autonomous decision-making capabilities. Unlike general digital tools that primarily enhance information availability or disclosure efficiency, AI technologies such as machine learning, natural language processing, knowledge representation and reasoning, and computer vision directly influence how information is processed, monitored, and acted upon. In doing so, AI constrains managerial discretion and affects substantive operational decisions rather than disclosure quality alone. While prior research shows that digital transformation including blockchain can enhance ESG performance (Jeong et al., 2025) and reduce ESG disclosure-performance misalignment (Chen et al., 2024), the specific role of AI (e.g., specific types) in this regard remains underexplored. Moreover, although existing evidence indicates that AI adoption improves ESG performance or ratings (S. Li et al., 2024; J. Li et al., 2025; Chen & Zhang, 2025; Lin & Zhu, 2025; Zhou et al., 2025), it remains unclear whether AI enhances the alignment between ESG disclosures and underlying practices. Addressing these gaps, this study conceptualizes AI as an internal

[5] Recent studies increasingly rely on AI-driven analytical tools to infer firms' ESG commitment and impression management by examining various features of ESG disclosures, including narrative sentiment (Lagasio, 2024), image usage (Albrecht et al., 2025) and visual prominence (Y. Li et al., 2025).

mechanism that disciplines ESG reporting by strengthening the linkage between ESG "talk" and ESG "action.".

2.3. Hypotheses Development

2.3.1. AI Utilization and ESG Disclosure-Performance Misalignment

We propose that AI utilization reduces ESG disclosure-performance misalignment through two interrelated but distinct mechanisms. The first mechanism operates through improvements in data processing and verification. AI enhances firms' information-processing and decision-making capabilities by enabling the rapid analysis of large, complex, and heterogeneous datasets, thereby improving data accuracy, completeness, and internal consistency (Commerford et al., 2022; Holmes & Douglass, 2022; Babina et al., 2024). In ESG contexts, relevant information is often fragmented across operational systems, supply chains, and external data sources, making manual aggregation and verification costly and error-prone. AI facilitates the continuous collection, integration, and cross-validation of ESG-related data from both internal operations and external sources, which substantially improves data quality and transparency (Agrawal et al., 2019; Chen et al., 2024).

By embedding algorithmic decision rules and automated analytics into organizational processes, AI standardizes data handling and verification procedures, reduces reliance on subjective managerial judgment, and enhances the consistency, reliability, and timeliness of ESG information processing (Kellogg et al., 2020). Moreover, AI-enabled analytics strengthen internal accountability by systematically linking disclosed ESG information to observable operational outcomes, thereby increasing the credibility of ESG reporting. For instance, by embedding advanced analytics into operational processes, AI allows firms to manage resources more efficiently, minimize waste, and maintain ongoing oversight of environmental compliance, thereby supporting more effective environmental management (Y. Liu et al., 2025). On the social front, AI applications help firms anticipate and mitigate workplace risks, identify safety lapses, and oversee labor practices throughout extended supply chains, enabling a more proactive approach to social responsibility (Agrawal et al., 2019; Helfat et al., 2023). In terms of governance, AI strengthens transparency and risk control by facilitating continuous assurance, timely detection of irregularities, and automated monitoring of fraudulent activities, which collectively enhance internal control systems and organizational accountability (Fedyk et al., 2022; Y. Liu et al., 2025). These illustrations suggest that higher-quality, timely and more verifiable ESG data enable firms to better manage ESG practices, thereby reducing the likelihood of ESG overstatement. Simultaneously, it also enables firms to more accurately identify, measure, and disclose genuine ESG initiatives, reducing ESG understatement.

The second mechanism operates through monitoring and governance. When monitoring is imperfect and costly, managers may exploit discretion to selectively disclose favourable ESG information while concealing adverse aspects of firm behaviour to manage investors, stakeholders and societal expectations (Delmas & Burbano, 2011; Walker & Wan, 2012). AI mitigates these frictions by reducing reliance on human judgment and enhancing the credibility of ESG information through automated monitoring, traceability, and anomaly detection (Kellogg et al., 2020; N. Li et al., 2024; S. Li et al., 2024). AI-enabled systems facilitate continuous and near real-time oversight of ESG-related activities, increasing the visibility of discrepancies between reported ESG claims and observed operational outcomes. This heightened visibility raises the expected costs of symbolic compliance and weakens managers' ability to sustain misaligned ESG narratives. By reducing information asymmetry (agency theory), improving firms' responsiveness to stakeholder demands (stakeholder theory), and increasing the costs of legitimacy-driven symbolic behaviour (legitimacy theory), AI adoption strengthens internal governance and transparency. Parallelly, enhanced monitoring improves internal coordination by ensuring that substantive ESG efforts are systematically recorded and reflected in disclosures. Collectively, these reduce ESG overstatement and understatement.

Accordingly, we hypothesize:

H1: *AI utilization is negatively associated with ESG disclosure-performance misalignment.*

2.3.2. AI Sub-types and ESG Disclosure-Performance Misalignment

Acknowledging that AI is not a homogeneous technology, we further explore how their impacts on ESG disclosure-performance misalignment vary substantially across underlying computational approaches. Prior literature distinguishes between logic-based AI and learning-based AI, which differ fundamentally in their reasoning processes, adaptability, data-processing capacity, and ability to operate under uncertainty (Townsend et al., 2025), as illustrated in Table 1. This distinction is particularly important in ESG settings, where information is heterogeneous, narrative-driven, and continuously evolving, and where AI effectiveness depends not only on rule compliance but also on detecting subtle misalignment between reported claims and underlying practices.

Learning-based AI, including machine learning (ML), natural language processing (NLP), and computer vision, relies on probabilistic and Bayesian reasoning to learn patterns from large volumes of structured and unstructured data (Townsend et al., 2025). By iteratively updating model parameters based on observed data, learning-based AI improves predictive accuracy over time and is capable of handling extraordinarily complex computational tasks (Agrawal et al., 2018). These capabilities make learning-based AI well suited to ESG contexts characterized by narrative disclosures, fragmented operational data, and external information from media, NGOs, and third-party ESG providers. Learning-based AI can detect inconsistencies, anomalies, and exaggerated sustainability narratives, for instance through NLP, while simultaneously integrating operational and environmental data to surface substantive but under-disclosed ESG activities (Sari et al., 2025).[6]

[6] However, learning-based systems are not without limitations; they may exhibit model brittleness when exposed to conditions far outside their training data and may generate opaque predictions that require careful governance (Mitchell, 2021).

Table 1
Comparisons of Logic-based AI and Learning-based AI.

	Learning-based AI	Logic-based AI
Programming Method and Logic	It is based on data-driven learning, whereby the system identifies features and patterns directly from large-scale datasets rather than operating on predefined rules. Model performance is improved through a formal training process in which parameters are iteratively optimized by algorithms, instead of being adjusted through manual coding.	It relies on explicit programming, where system behavior is governed by symbolic rules and logical propositions specified by human experts. The underlying rule set is finite in scope and implemented through manual coding.
Computational Reasoning	It is grounded in Bayesian or probabilistic reasoning, in which the system models the likelihood of events and produces predictive outputs that are inherently non-deterministic.	It utilizes propositional and deductive reasoning, operating within a predefined search space and producing deterministic outputs.
Ability to Process Knowledge	It is capable of extracting and leveraging tacit knowledge by emulating aspects of human perception. It identifies subtle and complex patterns in unstructured data, such as text and images through advanced pattern recognition techniques.	It is confined to explicit knowledge, requiring information to be formally represented, structured, and parameterized. Therefore, it is unable to process knowledge that cannot be readily articulated in symbolic or linguistic form.
Representative Technologies	Artificial neural networks Bayesian networks Deep learning	Expert systems Heuristic search Automated discovery

In contrast, logic-based AI relies on symbolic logic, predefined rules, and deterministic algorithms grounded in knowledge representation and reasoning (Townsend et al., 2025). Its reasoning is deductive, producing conclusions that are logically valid and explainable given explicitly coded assumptions and rules. Logic-based AI performs well in structured environments where problems can be represented as finite search spaces, making it effective for enforcing disclosure requirements, verifying consistency across ESG metrics, and generating auditable digital trails (Boden, 2014). However, because these systems depend on predefined rules and explicitly codified knowledge, they are inherently limited in adapting to evolving ESG standards, detecting nuanced or strategic misrepresentation, and processing large volumes of unstructured sustainability information. As complexity increases or new forms of ESG disclosure-performance misalignment emerge, computational demands can grow exponentially. Without human intervention, logic-based AI lacks the capacity to learn new possibilities in response such complexity (Boden, 2014; Chollet, 2019; Townsend et al., 2025).

These differences imply that learning-based AI is better positioned than logic-based AI to mitigate ESG disclosure-performance misalignment, which often manifests in subtle, evolving, and context-specific forms, such as inflated sustainability narratives, selective omission of negative outcomes, or delayed disclosure of substantive ESG actions. These forms extend beyond formal rule violations. Learning-based AI can process unstructured text, detect hidden patterns across multiple data sources, and adapt as ESG standards and disclosure practices evolve, thereby strengthening stakeholder scrutiny and increasing the costs of symbolic compliance. While logic-based AI supports *ex ante* compliance and standardization, learning-based AI more effectively constrains managerial discretion by uncovering discrepancies between ESG talk and action and by surfacing underreported ESG activities.

The preceding discussion leads us to hypothesize:

H2: *Learning-based AI adoption is negatively associated with ESG disclosure-performance misalignment, while the relationship between logic-based AI adoption and ESG disclosure-performance misalignment is weaker or insignificant.*

3. Research Design

3.1. Model Specification

Based on Zhou et al. (2025), we adopt the following Ordinary Least Squares model to examine the impact of AI technology utilization on ESG disclosure-performance misalignment:

$$ESG_DPM_{i,t} = \beta_0 + \beta_1 AI_UT_{i,t} + \beta_n \sum Controls_{i,t} + \sum Year + \sum Firm + \varepsilon_{i,t} \quad (1)$$

Where *ESG_DPM* represents the degree of misalignment between a firm's ESG disclosures and its actual ESG performance. *AI_UT* denotes the level of AI technology adoption within the firm, while *Controls* refer to a set of firm-level variables that may influence ESG disclosure-performance misalignment.

3.1.1. Dependent variable: ESG disclosure-performance misalignment

Following Gull et al. (2023a) and Aboud et al. (2024), we capture firms' actual ESG performance using ESG performance data provided by London Stock Exchange Group (LSEG), and firms' ESG disclosure quality with Bloomberg's ESG disclosure scores. Similar to Tashman et al. (2019), ESG disclosure-performance misalignment (*ESG_DPM*) is calculated by subtracting the standardized actual ESG performance score (*ESG_Action*) from the standardized ESG disclosure score (*ESG_Word*), as shown in Model (2).

$$ESG_DPM_{i,t} = \frac{ESG_{Word} - \overline{ESG}_{Word}}{\sigma_{Word}} - \frac{ESG_{Action} - \overline{ESG_{Action}}}{\sigma_{Action}} \quad (2)$$

We use Bloomberg's ESG Disclosure Score to capture *ESG_Word*. According to Bloomberg (2019), the Bloomberg ESG Disclosure

Score measures how comprehensively a firm reports its environmental, social, and governance information. The score reflects the transparency and breadth of publicly disclosed data. Bloomberg derives the score from more than 900 ESG-related fields collected from sustainability reports, annual reports, regulatory filings, company websites and other public disclosures (e.g., CDP data). These data points span key dimensions such as climate change, air quality, energy and water usage, waste management, labor practices, workplace diversity, human capital, executive compensation, board composition, shareholders' rights, and anti-corruption policies. Ranging from 0 to 100, a higher value of score indicates completer and more consistent ESG reporting. While useful for assessing ESG reporting practices, the metric does not capture the effectiveness or outcomes of a firm's ESG initiatives.

Next, we use LSEG's ESG score to capture a firm's *ESG_Action* in the main analyses. According to LSEG (2024), this score, ranging from 0 to 100, reflects a firm's relative ESG performance, commitment, and effectiveness using data-driven methodology. Drawing from more than 870 ESG metrics, LSEG evaluates firms across ten themes in the environmental (e.g., resource use, emissions, environmental innovation), social (e.g., workforce, human rights, community, product responsibility) and governance (e.g., management, shareholders, CSR strategy) pillars. Scores are benchmarked against industry peers (for environmental and social factors) and the country of incorporation (for governance), ensuring comparability across sectors and markets. The model incorporates industry-specific materiality weightings, recognising that the importance of ESG issues differs by sector. An ESG controversies overlay adjusts scores for verified negative events, with severity scaled to firm size. We triangulate our findings by employing the RepRisk Index from the RepRisk database as an alternative proxy for *ESG_Action* in robustness analysis.

To ensure comparability, *ESG_Word* and *ESG_Action* are standardized at the sample wide level prior to calculating *ESG_DPM*.[7] A positive value of *ESG_DPM* indicates greater ESG talk ("words") than ESG walk ("action"), and vice versa.

3.1.2. Test variable: AI technology utilization

Following the ML-based approach of Zhang and Yang (2024), we construct four categories of AI technology dictionaries to measure AI utilization within firms (*AI_UT*). Appendix 2 provides workflow charts detailing the procedures for classification determination and keyword selection across different AI types. The AI lexicon construction and measurement process involved the following steps:

First, firm annual reports were obtained from the Giant Tide Information Network and pre-processed by converting the documents into TXT format. Chinese word segmentation was performed using Jieba, with enhancements made through an AI-specific proper noun dictionary to improve accuracy.

Second, based on AI vocabulary frameworks developed by Deloitte (2020), iDigital (2023), and the International Monetary Fund (2023), we identified four core categories of AI technologies: ML, natural language processing (NLP), knowledge representation and reasoning, and computer vision. To ensure the representativeness and accuracy of the core seed words, we cross-validate and refine them based on the workflows illustrated in Appendix 2. Seed words from each category were used to train word vectors using the Word2Vec Skip-gram model, with Wikipedia data and a subset of annual reports forming the training corpus. The subset of annual reports for the training corpus was constructed using stratified random sampling, yielding 9,367 reports, which represent 20% of the full sample of 46,837 firm-year observations from 2011 to 2023. Using this representative corpus, we trained the Word2Vec (Skip-gram) model and expanded the initial seed words based on a cosine similarity threshold of 0.8. This threshold was selected through heuristic testing to balance semantic coverage and technical precision, thereby excluding generic business terms while retaining technology-specific identifiers (Song et al., 2025). For each seed word, the top 10 most semantically similar terms were selected based on cosine similarity scores. The AI technology dictionary was compiled after deduplication, manual verification, and semantic expansion.

To assess the validity of the keywords, we conducted a questionnaire survey (refer Appendix 6). A total of 150 questionnaires were distributed to industry practitioners, academic researchers, and government agencies. Out of these 137 were returned, yielding a response rate of 91.33%. After excluding 11 questionnaires with substantial missing responses and 8 exhibiting patterned or non-differentiated response behavior, the final sample comprised 118 valid questionnaires, representing 86.13% of the returned surveys.[8] Overall, the keyword lexicon designed to represent firms' AI technology applications was positively evaluated by experts, achieving an average score of 4.09 (out of 5). Most keywords attained mean scores above 4.0, indicating strong expert agreement regarding their accuracy and representativeness of underlying AI algorithms. Accordingly, keywords meeting this threshold were retained for the final seed vocabulary, while those lacking sufficient consensuses were excluded. Appendix 3 details the keywords for each category of AI.

To quantify the level of AI technology application, we calculate the frequency of AI-related keywords appearing in each firm's annual report and apply a log transformation (log(frequency + 1)) to construct *AI_UT*. A significantly negative coefficient on *AI_UT* would support Hypothesis 1, indicating that AI technology utilization helps mitigate ESG disclosure-performance misalignment.

Furthermore, we classify machine learning, natural language processing, and computer vision as learning-based AI technologies (*Learning*), while categorizing knowledge representation and logical reasoning as logic-based AI technologies (*KRR*). Such classifications are grounded in their core mechanisms, since *Learning* relies on learning latent data patterns to generate outputs, whereas *KRR* generates conclusions through logical deduction based on predefined facts and rules. For variable construction, a value of one is added

[7] Our results are robust to standardization conducted at the industry-year level (Section 4.4.1).

[8] Demographic statistics indicate that most respondents were affiliated with universities (41.53%) or enterprises (33.05%). Their primary areas of expertise included artificial intelligence research and development, data analytics and algorithm design, and academic research. 76 respondents (64.41%) were male and 42 (35.59%) were female. The majority of respondents fell within the 30–49 age group, accounting for 73.73% of the sample.

to the counts of each type of AI-technology keywords, followed by a logarithmic transformation, to derive the final measures for *Learning* and *KRR*. To support H2, we expect a significant and negative coefficient on *Learning*, but a weaker or even insignificant coefficient on *KRR*.

3.1.3. Control variables

We include a set of control variables related to firm and governance characteristics from that can influence ESG disclosure-performance misalignment (Chen et al., 2022; Liu et al., 2024; Gull et al., 2023a; Gull et al., 2023b). These data are collected from the China Stock Market & Accounting Research Database (CSMAR) database. The firm-level controls include firm size (*Size*), return on assets (*Roa*), Tobin's Q ratio (*TobinQ*), debt-to-asset ratio (*Lev*), capital intensity (*Cap*), cash flow ratio (*Cash*), revenue growth rate (*Growth*). Corporate governance variables include board size (*Board_Size*), proportion of independent directors (*Indep*), CEO-Chairperson duality (*Dual*), ownership concentration measured by the largest investor's shareholding (*Top1*) and institutional ownership (*INST*). The model includes year fixed effects to control for heterogeneity in ESG disclosure-performance misalignment over time, and firm fixed effects to eliminate time-invariant, unobservable firm characteristics that influence ESG disclosure-performance misalignment. Standard errors are clustered at both the firm and year levels (Petersen, 2009) to control for the within-firm and within-year correlations of ESG disclosure-performance misalignment. Appendix 1 details the measurement of all variables. Continuous variables are winsorized at the 1st and 99th percentiles to account for outliers.

3.2. Sample

We begin with Chinese A-share listed firms on the Shanghai and Shenzhen Stock Exchanges from 2011 to 2023. Panel A of Table 2 presents the sample selection procedure. The initial sample consists of 43,814 firm-year observations. First, following Long et al. (2025), we exclude 1,038 firm-year observations from financial, insurance, and investment industries. This is because they are differ systematically in: (i) business models and balance sheet structures (e.g., capital adequacy requirements and leverage definitions); (ii) regulatory oversight intensity and disclosure requirements; and (iii) ESG materiality profiles and measurement comparability, which jointly affect both the construction of our control variables and the interpretation of ESG measures.[9] We remove 1,042 firm-year observations with abnormal trading statuses (e.g., ST, *ST, and PT), followed by 18,430 firm-year observations with missing Bloomberg's ESG disclosure quality data, and 10,454 firm-year observations with missing LSEG's ESG performance data. The final sample comprises 12,850 firm-year observations from 1,369 firms spanning the 2011–2023 period. Panel B of Table 2 presents the mean comparisons of variables used in Model (1) between the final and excluded samples. The two key variables, namely *AI_UT* and *ESG_DPM*, show no statistically significant differences, indicating that sample selection does not bias the main constructs of interest.[10]

4. Main Results and Robustness Analyses

4.1. Descriptive Statistics

Table 3 presents the descriptive statistics of the key variables used in Model (1). The mean value of *ESG_DPM* is 0.412, with a minimum of −1.653 and a maximum of 2.235. These results suggest that, on average, Chinese listed firms exhibit significant misalignment between ESG disclosures and actual practices, with a general tendency to emphasize disclosure over substantive ESG practice (i.e., "talking more than doing"). *AI_UT* ranges from 0 to 1.609, reflecting uneven AI integration across firms. The descriptive statistics for control variables resemble those reported in prior Chinese studies (e.g., Long et al., 2025; Yang & Yang, 2025).

Table 4 presents the Pearson correlation matrix. *AI_UT* is significantly and negatively correlated with *ESG_DPM* (Correlation = -0.051), providing preliminary support to H1.

4.2. Main Results

Table 5 presents the main results. Column 1 includes only year and firm fixed effects, while Column 2 further includes control variables per Model (1). The coefficients on *AI_UT* are −0.111 in Column 1 and −0.101 in Column 2, respectively, and both are significant at the one percent level. This indicates that AI technology utilization is associated with lower levels of ESG disclosure-performance misalignment, thereby providing empirical support for H1.

We further assess whether learning-based AI technologies (*Learning*) and logic reasoning-based AI technologies (*KRR*) have differential impacts on ESG disclosure-performance misalignment. While incorporating these variables separately in the model, Column 3 shows that the negative coefficient of *Learning* (−0.107) is significant at the one percent level. However, Column 4 reveals that the negative coefficient (−0.87) on *KRR* is insignificant. Column 5 reports similar results when both *Learning* and *KRR* are incorporated simultaneously in the model. The results collectively validate H2.

This outcome can be attributed to two main factors. First, learning-based AI is more commonly adopted in corporate settings,

[9] Our findings are robust to the inclusion of firms from the financial, investment, and insurance industries as presented in Section 4.4.6.

[10] Several control variables, however, exhibit significant differences: firms in the final sample tend to be larger, more leveraged, hold more cash, but they also display lower growth opportunities and capital intensity compared to the excluded firms. Sample firms also have larger boards, less CEO-Chairperson duality, and higher largest concentrated and institutional ownership compared to excluded firms.

Table 2
Sample Selection and Characteristics.

Panel A: Sample selection procedures	
Data sample	Firm-year observations
Initial sample: Observations of Chinese A-share listed firms from fiscal years 2011–2023	43,814
Less: Observations from the financial, investment, and insurance industries	(1,038)
	42,276
Less: Observations of abnormal trading status (ST, *ST, PT companies)	(1,042)
	41,734
Less: Observations with missing Bloomberg's ESG disclosure quality data	(18,430)
	23,304
Less: Observations with missing LSEG's ESG performance data	(10,454)
Final number of observations	12,850

Panel B: Mean comparisons of Model (1) variables between the final versus excluded samples

Variable	(1) Mean of Final Sample	(2) Mean of Excluded Sample	*t*-value for the Mean Difference(1)-(2)
ESG_DPM	0.412	0.458	−0.942
AI_UT	0.077	0.076	0.300
Size	22.200	22.105	36.565***
Roa	0.037	0.038	−0.622
TobinQ	2.022	2.102	−2.608***
Lev	0.455	0.424	3.237***
Cap	2.297	14.329	−1.743**
Cash	0.053	0.040	13.168***
Growth	0.273	5.275	−0.736
Board Size	2.136	2.110	12.147***
Indep	0.377	0.376	1.385
Dual	0.251	0.316	−13.595***
Top1	0.341	0.334	4.939***
INST	0.461	0.423	14.327***

Table 3
Descriptive Statistics.

Variable	N	Mean	Std. Dev.	Min	Max
ESG_DPM	12,850	0.412	0.737	−1.653	2.235
AI_UT	12,850	0.071	0.266	0	1.609
Size	12,850	22.61	0.859	21.520	23.760
Roa	12,850	0.038	0.066	−0.225	0.232
TobinQ	12,850	1.953	1.255	0.810	8.095
Lev	12,850	0.455	0.199	0.066	0.897
Cap	12,850	2.217	1.621	0.378	9.908
Cash	12,850	0.053	0.066	−0.129	0.249
Growth	12,850	0.142	0.348	−0.498	2.062
Board Size	12,850	2.137	0.203	1.609	2.708
Indep	12,850	0.378	0.056	0.333	0.571
Dual	12,850	0.252	0.434	0	1
Top1	12,850	0.342	0.151	0.078	0.748
INST	12,850	0.461	0.238	0.006	0.902

Refer Appendix 1 for variable definitions.

whereas logic-based AI remains technically complex and less integrated into routine business operations (Suárez & Sánchez, 2024). Second, learning-based AI is highly adaptive, capable of processing large volumes of dynamic ESG data and continuously improving its analytical performance. This adaptability enhances the accuracy and reliability of ESG-related disclosures (Lin & Zhu, 2025). In contrast, logic-based AI relies on predefined rules and static inference structures, which limit its flexibility and effectiveness in handling the complexity and fluidity of ESG information.

4.3. Endogeneity Tests

4.3.1. Two-stage least squares (2SLS)

To address potential omitted variable bias, we employ an instrumental variable (IV) approach. Leveraging the Broadband China Strategy policy, we construct an instrument for firm-level AI technology utilization. Following Wang et al. (2022) and Peng et al. (2024), we use a dichotomous variable, *Broadband_China*, which equals one if a firm is headquartered in a city that has been designated as a "Broadband China" pilot city since 2014, and zero otherwise.

In theory, the designation of a city as a "Broadband China" pilot city should not directly affect firms' ESG disclosure performance.

Table 4
Correlation Matrix.

	ESG_DPM	AI_UT	*Size*	*Roa*	*TobinQ*	*Lev*	*Cap*	*Soe*	*Pollute*	$Board_Size$	*Indep*	*Dual*	*Top1*	*INST*
ESG_DPM	1													
AI_UT	−0.051***	1												
Size	0.155***	0.007	1											
Roa	0.073***	0.005	0.081***	1										
TobinQ	−0.039***	0.063***	−0.348***	0.227***	1									
Lev	0.082***	−0.060***	0.439***	−0.364***	−0.289***	1								
Cap	−0.069***	−0.006	−0.032***	−0.201***	−0.014	−0.102***	1							
Cash	0.041***	−0.015*	0.105***	0.459***	0.137***	−0.173***	−0.181***	1						
Growth	0.030***	−0.011	0.033***	0.296***	0.090***	−0.009	−0.109***	0.053***	1					
Board_Size	0.121***	−0.026***	0.246***	0.063***	−0.116***	0.109***	−0.010	0.054***	−0.005	1				
Indep	−0.024***	0.031***	0.013	−0.020**	0.042***	0.015*	0.014	0.007	−0.009	−0.517***	1			
Dual	−0.077***	0.060***	−0.184***	0.013	0.086***	−0.111***	0.022**	−0.022**	0.035***	−0.174***	0.098***	1		
Top1	0.164***	−0.113***	0.224***	0.161***	−0.095***	0.030***	−0.068***	0.125***	0.0120	0.075***	0.038***	−0.116***	1	
INST	0.163***	−0.097***	0.444***	0.161***	−0.048***	0.165***	−0.048***	0.150***	0.047***	0.233***	−0.016*	−0.222***	0.567***	1

See Appendix 1 for variable definition. *, **, and *** denote significance for the Pearson correlations at the 10, 5, and 1 percent levels, respectively.

Table 5
Main Results.

	(1)	(2)	(3)	(4)	(5)
	ESG_DPM	*ESG_DPM*	*ESG_DPM*	*ESG_DPM*	*ESG_DPM*
AI_UT	−0.111***	−0.101***			
	(−4.104)	(−3.759)			
Learning			−0.107***		−0.105***
			(−3.832)		(−3.730)
KRR				−0.087	−0.060
				(−1.441)	(−1.006)
Size		−0.173***	−0.172***	−0.176***	−0.172***
		(−11.415)	(−11.397)	(−11.610)	(−11.399)
Roa		0.160**	0.158**	0.171**	0.159**
		(2.029)	(2.007)	(2.169)	(2.016)
TobinQ		−0.013**	−0.013**	−0.013**	−0.013**
		(−2.537)	(−2.523)	(−2.506)	(−2.546)
Lev		0.314***	0.314***	0.315***	0.314***
		(7.589)	(7.586)	(7.598)	(7.586)
Cap		0.018***	0.018***	0.018***	0.018***
		(4.100)	(4.137)	(4.150)	(4.112)
Cash		−0.131*	−0.129*	−0.130*	−0.130*
		(−1.797)	(−1.771)	(−1.781)	(−1.791)
Growth		0.045***	0.045***	0.046***	0.045***
		(3.902)	(3.908)	(3.984)	(3.902)
Board_Size		−0.030	−0.030	−0.030	−0.030
		(−0.676)	(−0.690)	(−0.686)	(−0.688)
Indep		0.045	0.042	0.049	0.043
		(0.341)	(0.316)	(0.367)	(0.324)
Dual		−0.001	−0.001	−0.001	−0.001
		(−0.074)	(−0.088)	(−0.082)	(−0.086)
Top1		0.113	0.115	0.112	0.114
		(1.486)	(1.514)	(1.470)	(1.491)
INST		−0.151***	−0.151***	−0.148***	−0.152***
		(−3.061)	(−3.056)	(−2.982)	(−3.066)
Constant	0.420***	4.240***	4.232***	4.300***	4.234***
	(105.040)	(11.729)	(11.715)	(11.892)	(11.722)
Year FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
N	12,850	12,850	12,850	12,850	12,850
Adj.R^2	0.696	0.703	0.703	0.702	0.703

Columns 1 and 2 present the regression results of the impact of AI technology utilization level (*AI_UT*) on corporate ESG disclosure-performance misalignment (*ESG_DPM*), with versus without control variables, respectively. Columns 3 to 5 present the regression results based on different types of AI technologies: learning-based AI technology (*Learning*) and logic-based AI technology (*KRR*). Refer Appendix 1 for variable definition. Coefficient estimates are reported and the *t*-statistics are reported in parentheses. Standard errors are clustered at the firm and year levels. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

Although improvements in broadband infrastructure reduce information transmission costs and may facilitate activities such as remote monitoring or digital dissemination, broadband functions as a neutral technological conduit rather than a strategic determinant of disclosure behavior. The frequency and substance of ESG disclosures are primarily shaped by regulatory frameworks and managerial incentives, rather than by bandwidth constraints. Consequently, enhancements in network speed do not inherently motivate firms to misrepresent their ESG practices, thereby satisfying the exclusion restriction. However, such cities benefit from superior broadband infrastructure, which provides a more favourable environment for the adoption of AI technologies. This makes *Broadband_China* a theoretically sound and empirically valid instrumental variable, satisfying both the relevance (correlated with AI adoption) and exogeneity (unrelated to ESG disclosure-performance misalignment, except through AI adoption) conditions (Larcker & Rusticus, 2010).

Table 6 presents the results from the 2SLS regression analysis. The first-stage regresses *AI_UT* on *Broadband_China* and other control variables in Model (1). Column 1 indicates that the coefficient of the instrumental variable (*Broadband_China*) is positive (0.018) and significant at the five percent level, suggesting that the Broadband China Strategy policy designation drives AI technology adoption. The Kleibergen-Paap rk LM test (30.488) significantly rejects the null hypothesis of under-identification, confirming that the model does not suffer from identification issues. Furthermore, the Cragg-Donald Wald *F* statistic exceeds the critical threshold specified by the Stock-Yogo weak instrument test (29.428 > 16.380), indicating that *Broadband_China* is not a weak instrument. In the second-stage, we regress *ESG_DPM* on the predicted value of *AI_UT* (i.e., *Pred_AI_UT*) obtained from the first stage along with control variables in Model (1). *Pred_AI_UT* is significantly negative (−3.122) at the one percent level, reinforcing our main finding.

4.3.2. PSM-DID method

To further address sample selection bias and reverse causality, we employ a Propensity Score Matching (PSM)–Difference-in-

Table 6
Endogeneity Tests.

	First stage of 2SLS	Second stage of 2SLS	DID	PSM-DID
	(1)	(2)	(3)	(4)
	AI_UT	*ESG_DPM*	*ESG_DPM*	*ESG_DPM*
Broadband_China	0.018**			
	(2.428)			
Predicted_AI_UT		−3.122***		
		(−3.184)		
AIPZ			−0.166***	−0.084**
			(−9.768)	(−2.077)
Size	0.031***	−0.077**	−0.175***	−0.215***
	(5.114)	(−2.232)	(−11.656)	(−5.359)
Roa	−0.100**	−0.145	0.133*	0.448***
	(−2.349)	(−1.151)	(1.692)	(2.776)
TobinQ	−0.003	−0.023***	−0.012**	−0.002
	(−1.394)	(−3.741)	(−2.463)	(−0.156)
Lev	−0.006	0.299***	0.335***	0.530***
	(−0.360)	(7.186)	(8.094)	(5.222)
Cap	−0.004**	0.006	0.020***	0.016*
	(−2.320)	(1.100)	(4.707)	(1.769)
Cash	−0.032	−0.231***	−0.123*	−0.386**
	(−1.085)	(−2.877)	(−1.698)	(−2.335)
Growth	−0.011**	0.011	0.046***	0.062**
	(−2.414)	(0.709)	(3.948)	(2.373)
Board_Size	0.007	−0.011	−0.031	−0.073
	(0.341)	(−0.251)	(−0.719)	(−0.657)
Indep	−0.019	−0.020	0.032	0.313
	(−0.329)	(−0.148)	(0.244)	(0.923)
Dual	0.001	0.003	0.001	−0.003
	(0.223)	(0.268)	(0.084)	(−0.090)
Top1	−0.013	0.073	0.096	0.169
	(−0.563)	(0.934)	(1.267)	(0.947)
INST	−0.047**	−0.294***	−0.137***	−0.005
	(−2.443)	(−4.459)	(−2.771)	(−0.037)
Constant	−0.598***	2.435***	4.304***	4.928***
	(−4.213)	(3.540)	(12.005)	(5.348)
Kleibergen-Paap rk LM		30.488***		
Cragg-Donald Wald F		29.428***		
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
N	12,850	12,850	12,850	4,264
Adj.R^2	0.523	0.703	0.706	0.679

Column 1 presents the first-stage results of two stage least squares (2SLS) using *Broadband_China* as the instrumental variable. *Broadband_China* equals one if a firm is headquartered in a city designated as a Broadband China pilot city since 2014, and zero otherwise. Column 2 presents the second-stage results of 2SLS using the predicted value of *AI_UT* (*Pred_AI_UT*) obtained from the first stage as the test variable. Columns 3 and 4 present the results for the multi-period difference-in-differences analysis using the Artificial Intelligence Pilot Zone (*AIPZ*) policy as a quasi-natural experiment based on the full versus propensity score matched (PSM) samples, respectively. Refer Appendix 1 for variable definition. Coefficient estimates are reported and the *t*-statistics are reported in parentheses. Standard errors are clustered at the firm and year levels. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

Differences (DID) approach. Following Zhou et al. (2025), we leverage the designation of key cities under China's national-level *Artificial Intelligence Pilot Zone* (AIPZ) policy as a quasi-natural experiment. The AIPZ policy, launched in August 2019, aim to accelerate the integration of AI in the economic field, promote the development of innovative governance mechanisms in the intelligent era, and support the advancement of next-generation AI. By the end of 2021, approval had been granted in batches for 18 cities to carry out the construction of these pilot zones.[11]

The initiative of the National New-Generation Artificial Intelligence Innovation and Development Pilot Zones offers policy support through the provision of AI-related resources, real-world application scenarios, and advanced infrastructure. This policy has advanced the progress of regional AI technologies and created favourable conditions for promoting firm application of AI.

The multi-period Difference-in-Differences (DID) model is expressed in Model (3):

[11] These zones include: 2019- Beijing Municipality, Tianjin Municipality, Shanghai Municipality, Shenzhen City, Hangzhou City, Hefei City, Deqing County; 2020- Chongqing Municipality, Chengdu City, Xi'an City, Jinan City, Guangzhou City, Wuhan City; 2021- Suzhou City, Changsha City, Zhengzhou City, Shenyang City, Harbin City.

$$ESG_DPM_{i,t} = \partial_0 + \partial_1 AIPZ_{i,t} + \partial_n \sum Controls_{i,t} + \sum Year + \sum Firm + \varepsilon_{i,t} \quad (3)$$

If the firm is headquartered in a city included in the list of pilot policy in year *t*, then after year *t*, a value of one is assigned to *AIPZ*, and zero otherwise. For all other firms, it remains zero (Zhou et al., 2025). Column 3 of Table 6 presents that the coefficient of *AIPZ* is −0.166 which is significant at the one percent level, thereby supporting our main finding.

We apply this DID method using a propensity score matched (PSM) sample. To construct the PSM sample, we first regress *AIPZ* on the control variables specified in Model (1) to estimate propensity scores through a Logit regression. A 1:1 nearest neighbor matching approach with replacement is employed, using the specified control variables as matching criteria. Firms in the treatment group (*AIPZ = 1*) are then matched with those in the control group (*AIPZ = 0*) based on the closest propensity scores, thereby reducing systematic differences between groups and minimizing potential selection bias in the multi-period DID estimation. Using the PSM sample, Column 4 of Table 6 shows that the coefficient of *AIPZ* is negative (−0.084) and significant at the five percent level, supporting our main finding.

4.3.3. Parallel trend test

The validity of the DID approach relies on the parallel trends assumption, which requires that, prior to the implementation of the AIPZ policy, the ESG disclosure-performance misalignment trends between firms in pilot and non-pilot regions were statistically similar. Fig. 1 presents the results of the parallel trends test, using the policy implementation year as the reference period. The regression coefficients for all pre-policy periods are statistically insignificant and close to zero, with confidence intervals overlapping the zero line, indicating no significant differences in ESG disclosure-performance misalignment trends between the treatment and control groups prior to the policy. In contrast, significant divergence emerges after the policy implementation, reflecting the policy effect in constraining ESG disclosure-performance misalignment. These results confirm that the parallel trends assumption holds and further support the robustness of the DID regression findings discussed earlier.

4.3.4. Placebo test

To further verify the robustness of the DID results and account for potential unobservable factors, a placebo test was conducted. Specifically, 500 random samples were drawn to reconstruct the DID regression coefficients by randomly assigning treatment status to firms. As shown in Fig. 2, the distribution of the estimated placebo coefficients closely follows a standard normal distribution. The dashed line on the right indicates that the placebo coefficients are centered around zero. Importantly, the majority of these placebo coefficients are not statistically significant. Whereas, the solid line on the far left represents the average original *AIPZ* coefficient estimate (–0.166), which is statistically different from zero. These findings confirm that the observed treatment effect in the original DID analysis is unlikely to be driven by random chance, thereby reinforcing the statistical validity and reliability of the main results.

4.3.5. More versus less sustainable firms

Although we conduct several robustness checks including 2SLS and DID analyses to address potential reverse causality, and employ firm fixed effects to mitigate omitted variable concerns, the possibility that high-sustainability firms are inherently more likely to adopt AI cannot be fully ruled out. Therefore, we further estimate Model (1) within high- and low-sustainability subsamples, using green patent data and LSEG ESG scores, to assess whether the relationship remains consistent across different sustainability profiles. Specifically, green patents are identified using the International Patent Classification (IPC) Green Inventory developed by the World Intellectual Property Organization (WIPO). We partition the sample into high- and low-sustainability groups based on the industry-year median number of green patents. In addition, we employ ESG scores from LSEG, which provide a comprehensive measure of firms' overall ESG performance. Firms are similarly classified into high- and low-sustainability groups according to the industry-year median of LSEG ESG scores.

Results reported in Columns 1 to 4 of Table 7 show that *AI_UT* is significantly negative across all subsamples, although the magnitude of coefficient and significance are larger for the high-sustainability firms. Although these additional tests provide reassurance that our findings are not solely driven by pre-existing sustainability orientation, we acknowledge that causal inference in this setting remains subject to the usual limitations of observational data.

4.3.6. Lead-lag specification

Given the likelihood that the impact of AI technology utilization on corporate ESG disclosure-performance misalignment may take more than a year to materialize, and to address potential reverse causality, we re-estimate Model (1) with *AI_UT* and control variables that are lagged by one-period. Column 5 of Table 7 continues to support the baseline results, since lagged *AI_UT* yields a negative (−0.107) coefficient that is significant at the one percent level.

4.3.7. Additional control variables

Additionally, we extend Model (1) by incorporating industry- and region-level control variables, including the Herfindahl-Hirschman Index (HHI) to account for industry concentration and provincial GDP (*PGDP*) to capture regional economic development. The regression results are reported in Column 6 of Table 7. The coefficient on *HHI* is not statistically significant. Although *GDP* is significant, its coefficient is economically small in magnitude. The coefficient on *AI_UT* remains negative (−0.093) and significant at the one percent level, further corroborating the baseline findings.

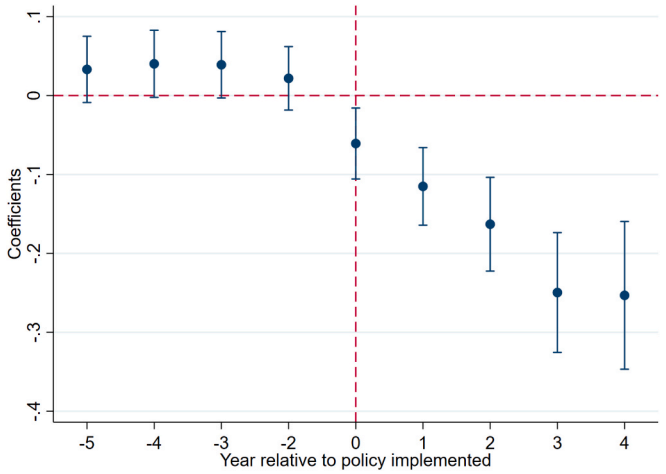


Fig. 1. Parallel trend test.

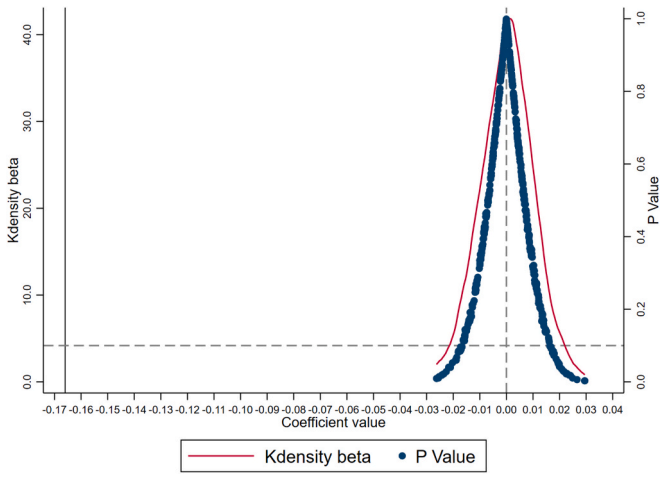


Fig. 2. Placebo test.

4.4. Other Robustness Tests

4.4.1. Alternative measures for ESG disclosure-performance misalignment and AI utilization

Berg et al. (2022) demonstrate that ESG rating agencies employ differing methodologies, with variations in measurement, scope, and weighting contributing to rating divergence. These factors influence inherent strengths and weaknesses of ESG ratings. For instance, LSEG provides comprehensive, industry-specific ESG assessments enriched by controversy analysis, but its reliance on firm disclosures may introduce reporting bias (Hinton, 2021; Lublóy et al., 2025). Whereas, Bloomberg's scores offer materiality-focused insights on ESG disclosures integrated with its financial data platform, but there are inconsistencies arising from non-standardized ESG reporting (Bloomberg, 2023; Lublóy et al., 2025).

Accordingly, it is essential to triangulate our results using ESG disclosure-performance misalignment measures derived from alternative rating providers. We use the RepRisk Index from the RepRisk database as a substitute indicator of actual ESG performance (*ESG_Action*) to compute ESG disclosure-performance misalignment, labelled *ESG_DPM_A1*. RepRisk (2021) captures ESG performance by systematically identifying and quantifying material ESG risks using an outside-in approach grounded in public, real-time information. Each day, RepRisk screens more than 150,000 sources in 23 languages, ranging from global and local media to regulators, NGOs, blogs, and social platforms, to detect ESG risk incidents linked to companies and projects. Its research framework is built around 28 core ESG Issues and 80 Topic Tags, mapped to 108 ESG risk factors and aligned with major international standards such as the Sustainable Development Goals, Sustainability Accounting Standards Board (SASB) Materiality Map, Organization for Economic Co-operation and Development (OECD) Guidelines, and global modern slavery legislation. Each incident is classified into one or more of these risk factors, providing a comprehensive picture of a firm's actual behavior. RepRisk then quantifies this exposure through two key metrics: the RepRisk Index (RRI), which ranges from 0 (lowest risk) to 100 (highest risk), and the RepRisk Rating (RRR), a letter

Table 7
Other Endogeneity Tests.

	Low-sustainability firms based on green patent	High-sustainability firms based on green patent	Low-sustainability firms based on LSEG's ESG performance score	High-sustainability firms based on LSEG's ESG performance score	Lead-lag	Additional controls
	(1)	(2)	(3)	(4)	(5)	(6)
	Dependent variable: *ESG_DPM*					
AI_UT	−0.085*	−0.123***	−0.058*	−0.095**	−0.107***	−0.093***
	(−1.678)	(−3.638)	(−1.806)	(−2.052)	(−3.763)	(−3.498)
PGDP						−0.001***
						(−7.531)
HHI						−0.003
						(−0.056)
Size	−0.221***	−0.102***	−0.250***	−0.048*	−0.218***	−0.174***
	(−9.734)	(−4.193)	(−12.013)	(−1.929)	(−13.499)	(−11.526)
Roa	0.291***	0.092	0.295***	0.175	0.012	0.125
	(2.705)	(0.786)	(3.123)	(1.329)	(0.142)	(1.596)
TobinQ	−0.025***	0.012	−0.010	−0.002	−0.059***	−0.012**
	(−3.924)	(1.396)	(−1.339)	(−0.252)	(−11.030)	(−2.458)
Lev	0.180***	0.555***	0.319***	0.272***	0.353***	0.320***
	(3.223)	(8.090)	(5.914)	(3.837)	(8.133)	(7.711)
Cap	0.010	0.017**	0.024***	0.007	0.015***	0.019***
	(1.615)	(2.434)	(4.847)	(0.827)	(3.235)	(4.394)
Cash	−0.091	−0.052	0.025	−0.223**	−0.090	−0.134*
	(−1.007)	(−0.452)	(0.255)	(−2.097)	(−1.244)	(−1.844)
Growth	0.048***	0.042**	0.071***	0.014	0.035***	0.045***
	(3.296)	(2.371)	(4.341)	(0.863)	(2.846)	(3.888)
Board_Size	−0.095	0.086	−0.067	0.007	−0.086*	−0.026
	(−1.578)	(1.310)	(−1.067)	(0.113)	(−1.918)	(−0.602)
Indep	−0.170	0.500**	−0.125	0.068	−0.026	0.028
	(−0.980)	(2.439)	(−0.648)	(0.345)	(−0.187)	(0.213)
Dual	0.007	0.026	0.018	−0.001	0.005	0.001
	(0.410)	(1.324)	(1.102)	(−0.073)	(0.371)	(0.094)
Top1	0.031	0.086	−0.210**	0.483***	0.087	0.088
	(0.291)	(0.728)	(−2.050)	(4.218)	(1.115)	(1.157)
INST	−0.132*	−0.158**	−0.063	−0.183**	−0.174***	−0.156***
	(−1.835)	(−2.115)	(−0.957)	(−2.349)	(−3.530)	(−3.137)
Constant	5.719***	2.002***	5.942***	1.414**	5.505***	4.565***
	(11.096)	(3.365)	(11.767)	(2.436)	(14.463)	(12.685)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
N	6,426	6,424	6,471	6,379	11,133	12,850
Adj.R^2	0.773	0.694	0.710	0.741	0.729	0.705

Columns 1 to 4 examine the impact of artificial intelligence (*AI_UT*) on ESG disclosure-performance misalignment (*ESG_DPM*) among firms with stronger versus weaker sustainability performance determined by industry-year median of green patent and LSEG's ESG performance scores, respectively. Column 5 presents the lead-lagged regression results, where *AI_UT* and control variables are lagged by one period. Column 6 presents results further controlling for industry concentration (*HHI*) and provincial GDP (*PGDP*). Refer Appendix 1 for variable definition. Coefficient estimates are reported and the *t*-statistics are reported in parentheses. Standard errors are clustered at the firm and year levels. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

grade from AAA to D, enabling clear benchmarking of ESG risk performance.[12]

Second, we re-estimate ESG disclosure-performance misalignment using ESG data standardized at the industry-year level, labelled as *ESG_DPM_A2,* since the original measure applied only sample wide level standardization.

Third, we redefine *AI_UT* to address potential measurement bias. In longer annual reports, the raw frequency of AI-related terms may be mechanically inflated, introducing noise into the measurement of AI technology utilization. In addition, this measure may be conflated with firms' general disclosure propensity and may exhibit mechanical correlation with the *ESG_DPM* construct, as both are partly derived from textual disclosures. To mitigate this, following Zhang and Yang (2024), the AI application variable is normalized and recalculated as the proportion of AI-related keywords relative to the total word count in each annual report, labelled as *AI_UTA1*.

Next, recognizing that keyword frequency in annual reports may capture rhetorical emphasis rather than genuine AI adoption, we triangulate this measure with alternative proxies of AI technology utilization to strengthen construct validity. Following J. Li et al. (2025), we use the number of a firm's AI-related technology patents and its AI-specific investments to capture substantive AI

[12] While RepRisk's strength lies in its ability to provide an independent, risk-focused, and broadly sourced assessment of ESG controversies (Lublóy et al., 2025), its limitation stems from a heavy reliance on external reporting, which can be backward-looking, as positive firm initiatives that may enhance future ESG performance often go unreported while external media tends to emphasize controversies and negative events (Barkemeyer et al., 2023).

engagement and to mitigate concerns about AI-washing. Data on AI patents and AI investments for listed companies are obtained from the CSMAR database. Our second alternative measure (*AI_UTA2*) is defined as the natural logarithm of the number of AI patents plus one, while the third measure (*AI_UTA3*) is calculated as the ratio of a firm's total AI-related investments to its total investments.

The results based on these alternative specifications, reported in Columns 1 through 5 of Table 8, remain fully consistent with our baseline findings.

4.4.2. Sample period adjustment

Since the breakthrough in deep learning in 2013, AI technology has advanced rapidly, with major global economies increasingly recognizing its strategic importance and transformative impact on socio-economic development. Hence, we restrict the sample period to the post-breakthrough years, covering 2014 to 2023, to better capture the developments in AI adoption and its potential effects. The negative coefficient on *AI_UT* (−0.084) with one percent level of significance, reported in Column 1 of Table 9, remains consistent with the baseline findings.

4.4.3. Controlling for digital transformation

While digital transformation captures a firm's broad adoption of digital infrastructure and applications (e.g., cloud computing, digital platforms, blockchain, smart manufacturing systems), AI represents a distinct, higher-order governance technology characterized by learning, prediction, and autonomous decision-making capabilities. Unlike general digital tools that primarily enhance information availability or disclosure efficiency, AI technologies (e.g., machine learning, natural language processing, knowledge representation and reasoning, and computer vision) directly affect how information is processed, monitored, and acted upon, thereby constraining managerial discretion and influencing real operational decisions rather than disclosure quality alone. Prior studies have shown that digital transformation contributes positively to corporate ESG performance and mitigates ESG disclosure-performance misalignment (Chen et al., 2024). To isolate the specific impact of AI technology utilization on ESG disclosure-performance misalignment, we control for the overall degree of corporate digital transformation (*Digital*) in Model (1).

Per Chen et al. (2024), the *Digital* variable is constructed through a textual analysis approach based on keywords detailed in Appendix 4. Using Python-based web scraping, we collect relevant textual data from firm disclosures. Based on four key dimensions, including digital technology applications, internet business models, smart manufacturing, and modern information systems, we tokenize the text and count keyword occurrences. These raw term frequencies are then normalized to construct the *Digital* index.

Column 2 of Table 9 reports the results, shows that without including *AI_UT* in the model, the coefficient of *Digital* is negative (−0.021) and significant at the one percent level, which is in line with Chen et al. (2024). Column 3 reveals that while *AI_UT* and *Digital* are simultaneously included in the model, both coefficients on *AI_UT* and *Digital* are negative and significant at the one percent level. Notably, the coefficient of *AI_UT* is −0.097, which is nearly 5.1 times larger of the coefficient of *Digital* (−0.019). This larger coefficient magnitude suggests that AI utilization provides incremental explanatory power beyond general digital transformation in reducing ESG disclosure-performance misalignment, offering additional insights beyond those reported by Chen et al. (2024).

4.4.4. Duration of implementation

To capture meaningful AI adoption and digital transformation, recognizing that benefits are likely to materialize as implementation matures, we construct *AI_UT_IMPL*, measured as the number of years since a firm first adopted AI technologies, and *Digital_IMPL*, measured as the number of years since a firm first adopted digital transformation initiatives. We then re-estimate Model (1) by replacing *AI_UT* with these two variables. The results reported in Column 4 of Table 9 suggest that the coefficient of *AI_UT_IMPL* is negative and significant (−0.024), and its magnitude is still larger than that of *Digital_IMPL* (−0.019).

4.4.5. Alternate fixed effects specification

We also adopt an alternative fixed-effects specification for Model (1) by replacing firm fixed effects with industry fixed effects while retaining year fixed effects. As shown in Column 5 of Table 9, *AI_UT* continues to exhibit a significantly negative association with *ESG_DPM* (−0.099), reinforcing the stability of our results.

4.4.6. Add samples from the financial sector

We expand the sample to include firms from the financial, insurance, and investment industries. As reported in Column 6 of Table 9, the coefficient on *AI_UT* remains significantly negative (−0.120), indicating that our findings are not sensitive to excluding financial firms.

5. Additional Analyses

5.1. Types of ESG Disclosure-Performance Misalignment

Since ESG disclosure-performance misalignment can manifest in two distinct forms, we analyse how AI technology utilization influences overstatement of ESG performance (*Talk* > *Action*) and understatement of ESG performance (*Action* > *Talk*). Appendix 5 provide instances of firms exhibiting these two types of ESG disclosure-performance misalignment. To execute this analysis and ease interpretation, *Talk* > *Action* is defined such that if *ESG_DPM* is greater than zero, its value is retained; otherwise, it is set to zero. Conversely, *Action* > *Talk* is defined such that if *ESG_DPM* is less than zero, its absolute value is retained; otherwise, it is set to zero. Column 1 of Table 10 shows that *AI_UT* is negative (–0.057) and significant at the one percent level, indicating that AI utilization

Table 8
Alternative Measures for ESG Disclosure-Performance Misalignment and AI Utilization.

	Alternative measure for ESG disclosure-performance misalignment		Alternative measure for AI utilization		
	(1)	(2)	(3)	(4)	(5)
	ESG_DPM_A1	*ESG_DPM_A2*	*ESG_DPM*	*ESG_DPM*	*ESG_DPM*
AI_UT	−0.146***	−0.161***			
	(−3.057)	(−3.179)			
AI_UTA1			−1.259**		
			(−2.156)		
AI_UTA2				−0.024**	
				(−2.561)	
AI_UTA3					−0.239**
					(−2.014)
Size	0.002	−0.316***	−0.175***	−0.174***	−0.175***
	(0.089)	(−11.589)	(−11.530)	(−11.481)	(−11.605)
Roa	−0.235*	0.043	0.167**	0.169**	0.155**
	(−1.836)	(0.290)	(2.116)	(2.142)	(1.968)
TobinQ	0.003	−0.108***	−0.013**	−0.013**	−0.012**
	(0.307)	(−11.479)	(−2.504)	(−2.491)	(−2.457)
Lev	0.197***	0.407***	0.314***	0.314***	0.314***
	(2.971)	(5.677)	(7.576)	(7.579)	(7.567)
Cap	0.034***	0.013*	0.018***	0.018***	0.018***
	(5.047)	(1.722)	(4.188)	(4.159)	(4.167)
Cash	−0.295**	−0.165	−0.130*	−0.130*	−0.125*
	(−2.451)	(−1.259)	(−1.781)	(−1.795)	(−1.717)
Growth	0.035*	0.019	0.046***	0.046***	0.046***
	(1.698)	(0.884)	(3.979)	(3.975)	(3.961)
Board_Size	−0.019	−0.116	−0.030	−0.031	−0.031
	(−0.213)	(−1.461)	(−0.688)	(−0.698)	(−0.707)
Indep	0.192	0.015	0.045	0.045	0.040
	(0.716)	(0.060)	(0.342)	(0.337)	(0.299)
Dual	−0.040*	0.004	−0.001	−0.001	−0.001
	(−1.717)	(0.175)	(−0.079)	(−0.054)	(−0.053)
Top1	0.204	0.145	0.116	0.116	0.107
	(1.469)	(1.078)	(1.522)	(1.517)	(1.406)
INST	−0.161**	−0.260***	−0.150***	−0.147***	−0.146***
	(−1.983)	(−3.007)	(−3.032)	(−2.979)	(−2.963)
Constant	0.254	7.481***	4.278***	4.275***	4.333***
	(0.379)	(11.386)	(11.825)	(11.804)	(11.942)
Year FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
N	12,850	12,850	12,850	12,850	12,850
Adj.R^2	0.518	0.598	0.703	0.703	0.703

Columns 1 to 5 present results using alternative measures of ESG disclosure-performance misalignment (*ESG_DPM_A1, ESG_DPM_A2*) and AI technology utilization (*AI_UTA1, AI_UTA2, AI_UTA3*), respectively. *ESG_DPM_A1* is estimated using the ESG reputation risk rating from RepRisk to proxy for ESG action. *ESG_DPM _A2* is estimated based on both ESG action and ESG talk being standardized at the industry-year levels. *AI_UTA1* is the proportion of AI-related keywords relative to the total word count in each annual report. *AI_UTA2* is the logarithm of a firm's number of AI technology patents plus one. *AI_UTA3* is calculated by the ratio of total AI investment to the company's total investment. Refer Appendix 1 for variable definition. Coefficient estimates are reported and the *t*-statistics are reported in parentheses. Standard errors are clustered at the firm and year levels. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

significantly reduces the overstatement of ESG performance. One possible reason is that AI utilization enhances the accuracy, traceability, and automation of ESG data collection and monitoring processes, making it more difficult for firms to overstate their ESG performance. Column 2 shows that *AI_UT* is also negatively (–0.043) and significantly associated with *Action* > *Talk* at the one percent level, suggesting that AI utilization reduces the understatement of ESG performance. One possible reason is that AI utilization improves firms' ability to capture, measure, and report positive sustainability outcomes that were previously overlooked or insufficiently documented, thereby boosting firm confidence and incentives to disclose such information. This ultimately reduces the understatement of ESG performance. Taken together, these results suggest that AI utilization mitigates both forms of ESG disclosure-performance misalignment.

5.2. Firm Heterogeneity

Firms of different sizes face varying levels of financial pressure, technological readiness, and access to skilled talent, all of which influence their ability to adopt AI technologies. Implementing AI often requires significant investment in infrastructure and expertise, posing greater challenges for smaller firms. In contrast, larger firms typically possess greater financial resources, a more advanced

Table 9
Other Robustness Tests.

	Reduced sample (2014 onwards)	Further controlling for digital transformation		Duration of Implementation	Industry fixed effects	Retain the financial sector sample
	(1)	(2)	(3)	(4)	(5)	(6)
	ESG_DPM	ESG_DPM	ESG_DPM	ESG_DPM	ESG_DPM	ESG_DPM
AI_UT	−0.084***		−0.097***		−0.099***	−0.120***
	(−2.935)		(−3.614)		(−3.716)	(−4.390)
Digital		−0.021***	−0.019***			
		(−3.560)	(−3.314)			
AI_UT_IMPL				−0.024***		
				(−4.135)		
Digital_IMPL				−0.019***		
				(−7.673)		
Size	−0.201***	−0.171***	−0.168***	−0.167***	−0.174***	−0.030***
	(−12.077)	(−11.192)	(−11.021)	(−11.046)	(−11.401)	(−2.582)
Roa	0.192**	0.169**	0.160**	0.123	0.154*	0.014
	(2.379)	(2.152)	(2.023)	(1.564)	(1.941)	(0.173)
TobinQ	−0.023***	−0.012**	−0.012**	−0.012**	−0.011**	0.001
	(−4.510)	(−2.415)	(−2.483)	(−2.504)	(−2.140)	(0.113)
Lev	0.402***	0.312***	0.312***	0.312***	0.310***	0.221***
	(8.878)	(7.535)	(7.531)	(7.547)	(7.422)	(5.260)
Cap	0.018***	0.018***	0.017***	0.017***	0.018***	0.002***
	(3.875)	(4.119)	(4.040)	(4.039)	(4.170)	(4.228)
Cash	−0.077	−0.132*	−0.135*	−0.126*	−0.147**	−0.155**
	(−1.006)	(−1.814)	(−1.854)	(−1.739)	(−2.019)	(−2.171)
Growth	0.054***	0.047***	0.046***	0.044***	0.047***	0.039***
	(4.205)	(4.037)	(3.943)	(3.829)	(4.043)	(3.451)
Board_Size	−0.033	−0.028	−0.027	−0.025	−0.035	0.072**
	(−0.677)	(−0.632)	(−0.624)	(−0.580)	(−0.798)	(2.008)
Indep	−0.072	0.032	0.031	0.034	0.072	0.006
	(−0.481)	(0.244)	(0.237)	(0.260)	(0.549)	(0.801)
Dual	0.003	−0.001	−0.001	−0.001	−0.002	0.011
	(0.204)	(−0.101)	(−0.089)	(−0.092)	(−0.166)	(0.867)
Top1	0.150*	0.108	0.107	0.102	0.120	0.304***
	(1.752)	(1.420)	(1.411)	(1.348)	(1.590)	(4.078)
INST	−0.227***	−0.146***	−0.151***	−0.169***	−0.161***	−0.242***
	(−4.020)	(−2.948)	(−3.042)	(−3.441)	(−3.312)	(−4.825)
Constant	4.880***	4.216***	4.164***	4.191***	4.267***	0.782***
	(12.327)	(11.609)	(11.469)	(11.627)	(11.684)	(3.280)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	No	Yes
Industry FE	No	No	No	No	Yes	No
N	11,727	12,850	12,850	12,850	12,850	13,888
Adj.R2	0.710	0.703	0.703	0.706	0.704	0.697

Column 1 presents results using a reduced sample from year 2014 onwards. Column 2 presents results removing *AI_UT* but incorporating digital transformation (*Digital*). Column 3 presents results when both *AI_UT* and *Digital* are simultaneously incorporated in the model. Column 4 presents the results on the effects of the years since AI implementation (*AI_UT_IMPL*) and the years since digital transformation (*Digital_IMPL*). Column 5 presents the regression results replacing firm fixed effects with industry fixed effects. Column 6 presents the regression results by including firms from the financial, investment and insurance industries. Refer Appendix 1 for variable definition. Coefficient estimates are reported and the *t*-statistics are reported in parentheses. Standard errors are clustered at the firm and year levels. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

technological foundation, and a deeper talent pool, making them better equipped to implement and benefit from AI. This enhanced capacity enables them to more effectively improve the quality and reliability of ESG disclosures. Accordingly, we examine whether the negative relationship between *AI_UT* and *ESG_DPM* varies with firm size. We split firms into two subgroups (large versus small) based on the industry-year median firm size. As shown in Columns 1 and 2 of Panel A in Table 11, the coefficient on *AI_UT* is negative (−0.178) and significant at the one percent level among larger firms, whereas the positive coefficient (0.001) on *AI_UT* is statistically insignificant for smaller firms.[13] Columns 3 and 4 exhibit consistent patterns when *ESG_DPM_A2* is used as the dependent variable. These findings reveal an uneven landscape of technology-enabled governance, where larger firms are structurally better equipped to

[13] We analyze the confidence interval of *AI_UT* for smaller firms. The interval width is 0.0394261 − (−0.0245759) = 0.064002, and its central point is (−0.0245759 + 0.0394261) / 2 = 0.0074251. The confidence interval is relatively narrow and centered close to zero, indicating adequate sample size and high estimation precision. This evidence supports the conclusion that the application of AI has no statistically significant impact on smaller firms.

Table 10
Additional Analyses: Type of ESG Disclosure-Performance Misalignment.

Variables	(1)	(2)
	Talk > Action	*Action > Talk*
AI_UT	−0.057***	−0.043***
	(−3.129)	(−2.811)
Size	−0.087***	−0.086***
	(−7.399)	(−10.616)
Roa	0.082	0.078*
	(1.439)	(1.788)
TobinQ	−0.009**	−0.004
	(−2.341)	(−1.370)
Lev	0.159***	0.156***
	(4.954)	(8.141)
Cap	0.011***	0.006***
	(3.661)	(2.734)
Cash	−0.097*	−0.034
	(−1.756)	(−0.933)
Growth	0.018**	0.027***
	(2.101)	(4.619)
Board_Size	−0.059*	0.030
	(−1.722)	(1.524)
Indep	0.025	0.020
	(0.256)	(0.321)
Dual	−0.005	0.004
	(−0.562)	(0.672)
Top_1	0.009	0.105***
	(0.142)	(3.071)
INST	−0.095**	−0.057**
	(−2.472)	(−2.391)
Constant	2.523***	1.717***
	(9.004)	(8.986)
Year FE	Yes	Yes
Firm FE	Yes	Yes
N	12,850	12,850
Adj.R2	0.781	0.414

This table presents the analysis based on the types of ESG disclosure-performance misalignment, including talks outweigh actions (*Talk > Action*) and actions outweigh talks (*Action > Talk*). Appendix 5 provide anecdotal illustrations of firms engaging in overclaiming and underclaiming their ESG performance. Refer Appendix 1 for variable definition. Coefficient estimates are reported and the *t*-statistics are reported in parentheses. Standard errors are clustered at the firm and year levels. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

deploy AI for credible ESG verification. This advantage enable them to more effectively align ESG practices with their disclosures. As capital markets increasingly reward data-driven credibility, firm size emerges as a key factor in determining whether AI adoption leads to greater investor trust and improved access to sustainable capital. From a practical perspective, regulators could mitigate this imbalance by promoting shared AI infrastructure, AI-enabled ESG reporting tools, or targeted incentives that lower the cost of AI adoption for smaller firms.

The effectiveness of AI in mitigating ESG disclosure-performance misalignment may also vary between state-owned enterprises (SOEs) and non-SOEs. We thus re-estimate Model (1) in SOE versus non-SOE subsamples. Columns 5 and 6 of Panel A in Table 11 show that the coefficient of *AI_UT* is negative for both non-SOEs and SOEs, however the magnitude of coefficient and level of significance are greater for non-SOEs (−0.103, one percent significance) relative to SOEs (−0.073, non-significance).[14] One possible explanation is that SOEs often operate within more complex bureaucratic structures and under stricter information controls. Powerful control by the governments can hinder the flexible application of AI and reduce data timeliness and adaptability. In contrast, non-SOEs typically exhibit more agile organizational structures and are more responsive to technological innovation, facilitating more efficient and effective integration of AI tools. As a result, non-SOEs experience a stronger reduction in ESG disclosure-performance misalignment driven by AI utilization. Columns 7 and 8 yield consistent results when *ESG_DPM_A2* is employed as the dependent variable. The stronger impact of AI on reducing the ESG disclosure-performance misalignment among non-SOEs suggests that governance value of AI depends critically on institutional compatibility. Non-SOEs can leverage organizational agility to rapidly translate AI insights into higher disclosure quality and better decision-making. In contrast, the multi-layered decision hierarchies in SOEs seem to hinder this

[14] We examine the confidence interval of *AI_UT* for SOEs. The width of this interval is calculated as 0.0178065 − (−0.1638647) = 0.1816712, and its central point is (−0.1638647 + 0.0178065) / 2 = −0.0730291. The confidence interval is relatively wide, and its midpoint is not tightly centered around zero. This suggests that the estimate for SOEs should be interpreted with caution in relation to that of non-SOEs.

Table 11
Additional Analyses: Firm heterogeneity and Industry heterogeneity.

Panel A: Firm heterogeneity

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Larger firms	Smaller firms	Larger firms	Smaller firms	SOE	Non-SOE	SOE	Non-SOE
Variables	*ESG_DPM*	*ESG_DPM*	*ESG_DPM_A2*	*ESG_DPM_A2*	*ESG_DPM*	*ESG_DPM*	*ESG_DPM_A2*	*ESG_DPM_A2*
AI_UT	−0.178***	0.007	−0.263***	0.026	−0.073	−0.103***	−0.027	−0.201***
	(−4.256)	(0.455)	(−3.290)	(0.854)	(−1.576)	(−3.164)	(−0.305)	(−3.284)
Constant	−0.181	3.307***	1.908	5.682***	−0.235	6.438***	−0.493	13.051***
	(−0.265)	(6.608)	(1.485)	(6.734)	(−0.460)	(13.740)	(−0.531)	(14.834)
Controls, Year FE, Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj.R2	0.739	0.848	0.663	0.779	0.729	0.680	0.634	0.568
N	6,150	6,700	6,150	6,700	5,285	7,565	5,285	7,565

Panel B: Industry heterogeneity

Variables	(1) Capital-intensive industry	(2) Labor-intensive industry	(3) Technology-intensive industry	(4) Capital-intensive industry	(5) Labor-intensive industry	(6) Technology-intensive industry
Variables	*ESG_DPM*	*ESG_DPM*	*ESG_DPM*	*ESG_DPM_A2*	*ESG_DPM_A2*	*ESG_DPM_A2*
AI_UT	−0.282**	0.072	−0.087***	−0.391*	0.169	−0.147***
	(−2.570)	(0.981)	(−2.924)	(−1.840)	(1.071)	(−2.675)
Constant	0.981	4.511***	6.334***	1.819	8.286***	11.034***
	(1.396)	(6.087)	(10.512)	(1.416)	(6.161)	(9.983)
Controls, Year FE, Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
N	3,991	3,644	5,215	3,991	3,644	5,215
Adj.R^2	0.731	0.729	0.678	0.642	0.605	0.566

Panel A presents cross-sectional analyses based on firm size and state-ownership. Panel B presents industry heterogeneity analysis, where firms are split into capital-intensive, labor-intensive, and technology-intensive industries. Refer Appendix 1 for variable definition. Coefficient estimates are reported and the *t*-statistics are reported in parentheses. Standard errors are clustered at the firm and year levels. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

process. From a practical perspective, this suggests that unlocking AI's potential within SOEs necessitates adjustments in governance such as streamlining decision-making and approval processes, rather than relying solely on the technological adoption.

5.3. Industry Heterogeneity

Given the variation in AI utilization across industries, ranging from differences in technology types and application intensity to resource availability, we further classify firms into three groups based on the CSRC's 2012 industry classification: technology-intensive, capital-intensive, and labour-intensive sectors. As shown in Columns 1–3 of Panel B, Table 11, the negative relationship between *AI_UT* and *ESG_DPM* holds in capital-intensive industries (−0.282, five percent significance) and technology-intensive industries (−0.087, one percent significance). These industries are more reliant on advanced technologies and specialized expertise, and they typically possess more robust technological infrastructure and well-developed data management systems. Such characteristics enhance their ability to integrate AI effectively into ESG reporting processes. In contrast, labour-intensive industries, which depend more on manual operations and often allocate fewer resources to technology advancement, derive comparatively limited benefits from AI in this context. We observe consistent results in Columns 4–6 when *ESG_DPM_A2* is used as the dependent variable. The industry-level heterogeneity suggests that the governance value of AI depends on how ESG outcomes are produced within different sectors. In capital- and technology-intensive industries, ESG performance is closely embedded in production systems, equipment, and processes. This results in generating standardized and traceable data (e.g., energy use and emission intensity) that AI can readily monitor and reconcile with disclosures. In contrast, in labor-intensive industries, ESG outcomes are largely driven by workforce practices, organizational culture, and interpersonal interactions, which are less standardized and harder to codify, thereby limiting the scope for AI-based verification. These findings imply that AI is most effective at reducing ESG disclosure-performance misalignment in industries where sustainability outcomes are operationally integrated and generated by system, revealing an industry boundary for the governance role of AI.

5.4. Economic Consequences

China has been intensifying its regulatory oversight of green finance, notably through the issuance of the Green Credit Guidelines by the China Banking Regulatory Commission (CBRC) in 2012. These developments underscore the country's commitment to channelling financial resources toward genuinely sustainable businesses (Tian et al., 2024). However, this has raised the stakes for firms that exhibit deviations between ESG disclosure and performance. ESG disclosure-performance misalignment may signal to the market and regulators that a firm's green credentials are superficial, thereby undermining trust and heightening green financing constraints.

The utilization of AI may serve as a strategic governance tool to enhance the authenticity of ESG practices and reduce ESG risks. Reduced ESG disclosure-performance misalignment, in turn, strengthens firms' positioning in the green credit market.

Building on Xing et al. (2021), we measure firms' access to green credit (*GC*) using the ratio of green credit to total loans, where a higher ratio indicates stronger access to green finance and lower financing constraints. We conduct path analysis by estimating a structural equation model (SEM) that examines the direct impact of *AI_UT* on *GC*, while using *ESG_DPM* as a mediating variable to demonstrate its indirect impact. Table 12 presents our regression results. Columns 1 and 3 show that the direct effect of *AI_UT* on *GC* (0.003 and 0.002, respectively) is significant, suggesting that firms with greater AI utilization have stronger access to green credit. Column 2 shows that *AI_UT* is negatively and significantly associated with *ESG_DPM* (−0.101), which is consistent with our main finding. Column 3 shows that the coefficient of *ESG_DPM* on *GC* is negative (−0.004) and significant. These give rise to an indirect coefficient for *AI_UT* on *GC* equal to 0.0003, with a significant *z*-statistic of 3.34. Therefore, the results support the indirect effect, depicting that AI utilization helps firms obtain more green credit by reducing ESG disclosure-performance misalignment.

6. Conclusion

Using China as an empirical context, this study finds that the utilization of AI technology mitigates ESG disclosure-performance misalignment, including both forms of overstatement and understatement of ESG performance. Our results are robust to various robustness and endogeneity tests. This includes two-stage least squares, propensity score matching combined with difference-in-differences, parallel trend tests, placebo tests, and lead-lag specifications. The results also hold under alternative measures of ESG

Table 12
Economic Consequences: Green Credit.

	(1)	(2)	(3)
	GC	*ESG_DPM*	*GC*
AI_UT	0.003***	−0.101***	0.002**
	(2.850)	(−3.759)	(2.463)
ESG_DPM			−0.004***
			(−7.079)
Size	−0.004***	−0.173***	−0.005***
	(−5.980)	(−11.415)	(−6.874)
Roa	−0.004	0.160**	−0.004
	(−1.193)	(2.029)	(−1.033)
TobinQ	−0.000**	−0.013**	−0.001**
	(−2.140)	(−2.537)	(−2.348)
Lev	−0.003*	0.314***	−0.002
	(−1.720)	(7.589)	(−1.100)
Cap	−0.001**	0.018***	−0.000**
	(−2.574)	(4.100)	(−2.265)
Cash	−0.006*	−0.131*	−0.007**
	(−1.883)	(−1.797)	(−2.034)
Growth	0.000	0.045***	0.000
	(0.495)	(3.902)	(0.792)
Board_Size	−0.001	−0.030	−0.001
	(−0.726)	(−0.676)	(−0.786)
Indep	−0.014**	0.045	−0.014**
	(−2.540)	(0.341)	(−2.515)
Dual	−0.001	−0.001	−0.001
	(−1.563)	(−0.074)	(−1.566)
Top1	0.009***	0.113	0.009***
	(2.731)	(1.486)	(2.854)
INST	−0.005**	−0.151***	−0.005**
	(−2.166)	(−3.061)	(−2.416)
Constant	0.163***	4.240***	0.179***
	(9.968)	(11.729)	(10.890)
Year FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
N	12,850	12,850	12,850
Adj.R^2	0.802	0.703	0.804
Mediating effects			
Indirect effect: *AI_UT* × *ESG_DPM*	0.0003***		
z-statistic for indirect effect: *AI_UT* × *ESG_DPM*	(3.340)		
Direct effect:	0.0023		
Total effect:	0.0027		
% of the total mediated effect:	13.45%		

This table presents the economic benefits captured by the firm's access to green credit (*GC*) through which AI technology utilization (*AI_UT*) reduces ESG disclosure-performance misalignment (*ESG_DPM*) based on a structural equation model. Refer Appendix 1 for variable definition. Coefficient estimates are reported and the *t*-statistics are reported in parentheses. Standard errors are clustered at the firm and year levels. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

disclosure-performance misalignment. Specifically, we reconstruct ESG disclosure-performance misalignment using ESG reputation risk from RepRisk as a proxy for ESG actions and employ industry-year standardized ESG disclosure-performance misalignment measures based on Bloomberg and LSEG data. Similarly, our conclusions remain unchanged when AI utilization is measured using alternative proxies, including: (i) scaling AI-related keywords by total word count in annual reports to control for disclosure propensity; (ii) using years since AI implementation to capture lagged realized benefits of AI adoption; and (iii) addressing concerns about AI-washing by incorporating more substantive indicators such as AI-related patent counts and AI investment data. We further find that learning-based AI effectively curbs ESG disclosure-performance misalignment compared to logic-based AI. Moreover, the benefits of AI in reducing ESG disclosure-performance misalignment are more evident in larger firms, non-state-owned firms and firms operating in capital- and technology-intensive industries. Finally, we document that reduced AI utilization is associated with a greater likelihood of firm's access to green credit, mediated through reduced ESG disclosure-performance misalignment.

This study offers several practical implications. First, regulators should expand access to high-quality information resources and support open, industry-wide data platforms to strengthen AI's governance and informational role in curbing ESG disclosure-performance misalignment. Cross-sector data sharing and stable, long-term policy frameworks that encourage AI research and deployment are critical to enabling effective AI integration into corporate ESG governance. Importantly, our findings show that not all forms of AI are equally effective. Learning-based AI, characterized by adaptive algorithms, predictive analytics, and continuous learning, substantially outperforms logic-based AI in narrowing the gap between disclosed and actual ESG performance. This underscores the need to move beyond generic digital adoption toward targeted investment in advanced AI systems that enhance data integrity, verification, and real-time oversight. Regulators could employ AI-enabled text analytics and anomaly detection to identify ESG misalignment and disclosure risk, while firms could deploy tools such as NLP-based consistency checks, machine-learning models for ESG risk monitoring, and AI-assisted internal controls to strengthen the alignment between ESG narratives and underlying actions.

This study has several limitations. First, our primary measure of AI utilization relies on the frequency of AI-related keywords appear in firms' annual reports. However, this approach may not fully capture the depth, sophistication, or operational effectiveness of AI implementation within firms. Keyword-based measures are inherently limited in their ability to distinguish between symbolic disclosure and substantive technological deployment. As a result, they may partially represent broader disclosure practices, communication strategies, or narrative styles rather than actual AI capabilities. In addition, textual classifications may not fully capture heterogeneity in AI maturity, scale, or integration across different business functions, nor can they perfectly differentiate between experimental, pilot-stage, and enterprise-wide AI applications.

To address these concerns, we use alternative proxies to triangulate AI utilization and by validating our keyword lexicon through expert assessment. Future research could further enhance measurement precision by adopting complementary approaches. These may include firm-level surveys of AI deployment, validated AI readiness or digital maturity indices, patent- or investment-based measures of AI capabilities, or more granular and standardised disclosures on technology investments across different AI categories. Advances in data availability and regulatory guidance on digital and AI-related reporting may also enable future studies to more accurately capture the intensity and effectiveness of AI adoption at the firm level.

Second, although ESG disclosure-performance misalignment is constructed using reputable third-party databases, differences in data construction, weighting schemes, and coverage across ESG rating agencies may introduce measurement noise and limit direct comparability. Future studies could address this issue by employing multi-rater consensus measures, rater-specific fixed effects, or latent-variable approaches to better isolate the ESG constructs. Moreover, we acknowledge that our *ESG_DPM* measure captures the degree and direction of misalignment between ESG disclosures and underlying ESG performance, rather than intentional deception per se. Future research could explicitly examine extreme cases of greenwashing or other distinct forms of ESG decoupling.

Third, we acknowledge the limits of causal inference despite employing firm fixed effects, instrumental variable approach, DID designs, and extensive robustness checks. We clarify that these approaches strengthen internal validity but do not eliminate all concerns related to unobserved time-varying factors or concurrent policy and technology shocks. We therefore frame our results as providing strong evidence consistent with a causal interpretation, while encouraging future research to test these mechanisms in alternative institutional settings and with complementary research designs.

Lastly, this study focuses exclusively on Chinese A-share listed firms, which may limit the generalisability of our findings given China's distinctive institutional, regulatory, and disclosure environment. Comparative analyses across economies with different regulatory environments, sustainability norms, and AI adoption trajectories would provide richer insights into the global applicability and effectiveness of AI in strengthening ESG governance.

Declaration of Generative AI in Scientific Writing: Generative AI and AI-assisted technologies were only used in the writing process to improve the readability and language of the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This research received the financial support from National Social Science Fund of China Project, Grant/Award Number: 24FGLB166.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.accinf.2026.100773.

Data availability

Data that support the findings of this study are from paid subscriptions. Restrictions apply to the availability of these data, which were used under license for this study.

References

Aboud, A., Saleh, A., Eliwa, Y., 2024. Does mandating ESG reporting reduce ESG decoupling? evidence from the European Union's Directive 2014/95. Bus. Strateg. Environ. 33 (2), 1305–1320.

Agrawal, A., Gans, J., Goldfarb, A., 2018. *Prediction machines: the simple economics of artificial intelligence*. Harvard Business Press.

Agrawal, A., Gans, J.S., Goldfarb, A., 2019. Artificial intelligence: the ambiguous labor market impact of automating prediction. J. Econ. Perspect. 33 (2), 31–50.

Albrecht, D.H., Anantharaman, D., Zhao, K., 2025. Is a picture worth a thousand words? image usage in ESG reports. Acc. Organ. Soc. 115, 101616.

Babina, T., Fedyk, A., He, A., Hodson, J., 2024. Artificial intelligence, firm growth, and product innovation. J. Financ. Econ. 151, 103745.

Barkemeyer, R., Revelli, C., Douaud, A., 2023. Selection bias in ESG controversies as a risk for sustainable investors. J. Clean. Prod. 405, 137035.

Berg, F., Kölbel, J.F., Rigobon, R., 2022. Aggregate confusion: the divergence of ESG ratings. Eur. Finan. Rev. 26 (6), 1315–1344.

Bloomberg. (2019). *Environmental, social & governance (ESG) product*. Bloomberg L.P.

Bloomberg. (2023). *Bloomberg ESG Scores*. https://hr.bloombergadria.com/data/files/Pitanja%20i%20odgovori%20o%20Bloomberg%20ESG%20Scoreu.pdf#:~:text=The%20scores%20are%20based%20on%20publicly%20available%2C,quantitative%20data%20as%20a%20dimension%20of%20performance.

Boden, M.A., 2014. GOFAI. In: Frankish, K., Ramsey, W. (Eds.), The Cambridge Handbook of Artificial Intelligence. Cambridge University Press, Cambridge, U.K., pp. 89–107

Boritz, J.E., Stratopoulos, T.C., 2023. AI and the accounting profession: views from industry and academia. J. Inf. Syst. 37 (3), 1–9.

Brynjolfsson, E., Hui, X., Liu, M., 2019. Does machine translation affect international trade? evidence from a large digital platform. Manag. Sci. 65 (12), 5449–5460.

Carlos, W.C., Lewis, B.W., 2018. Strategic silence: Withholding certification status as a hypocrisy avoidance tactic. Adm. Sci. Q. 63 (1), 130–169.

Chen, P., Gao, J., Ji, Z., Liang, H., Peng, Y., 2022. Do artificial intelligence applications affect carbon emission performance?—evidence from panel data analysis of chinese cities. Energies 15 (15), 5730.

Chen, R., Zhang, T., 2025. Artificial intelligence applications implication for ESG performance: can digital transformation of enterprises promote sustainable development? Chin. Manag. Stud. 19 (3), 676–701.

Chen, X., Wan, P., Ma, Z., Yang, Y., 2024. Does corporate digital transformation restrain ESG decoupling? evidence from China. Humanities and Social Sciences Communications 11 (1), 1–15.

Chollet, F., 2019. *On the measure of intelligence*. arXIV, Cornell University, Ithaca, NY.

Christ, M.H., Emett, S.A., Summers, S.L., Wood, D.A., 2021. Prepare for takeoff: improving asset measurement and audit quality with drone-enabled inventory audit procedures. Rev. Acc. Stud. 26 (4), 1323–1343.

Christensen, H.B., Hail, L., Leuz, C., 2021. Mandatory CSR and sustainability reporting: economic analysis and literature review. Rev. Acc. Stud. 26 (3), 1176–1248.

Commerford, B.P., Dennis, S.A., Joe, J.R., Ulla, J.W., 2022. Man versus machine: complex estimates and auditor reliance on artificial intelligence. J. Account. Res. 60 (1), 171–201.

De Villiers, C., Marques, A., 2016. Corporate social responsibility, country-level predispositions, and the consequences of choosing a level of disclosure. Account. Bus. Res. 46 (2), 167–195.

Deegan, C., 2002. Introduction: the legitimising effect of social and environmental disclosures–a theoretical foundation. Account. Audit. Account. J. 15 (3), 282–311.

Delmas, M.A., Burbano, V.C., 2011. The drivers of greenwashing. Calif. Manage. Rev. 54 (1), 64–87.

Deloitte. (2020). *Winning in the Age of Enterprise Intelligence: A Survey of Artificial Intelligence Application Practices among Chinese and Global Enterprises*. https://stock.tianyancha.com/qmp/report/2/20882cee9a0cfaae48713c5784cec703.pdf.

Ding, K., Lev, B., Peng, X., Sun, T., Vasarhelyi, M.A., 2020. Machine learning improves accounting estimates: evidence from insurance payments. Rev. Acc. Stud. 25 (3), 1098–1134.

Du, X., 2015. How the market values greenwashing? evidence from China. J. Bus. Ethics 128 (3), 547–574.

Eliwa, Y., Elmaghrabi, M.E., 2025. Investment horizons and ESG decoupling: distinct roles of long-term and short-term institutional investors. Econ. Lett. 247, 112207.

Fedyk, A., Hodson, J., Khimich, N., Fedyk, T., 2022. Is artificial intelligence improving the audit process? Rev. Acc. Stud. 27 (3), 938–985.

Freeman, R.E., 1984. *Strategic management: a stakeholder approach*. Pitman, Boston.

García-Sánchez, I.-M., Hussain, N., Khan, S.-A., Martínez-Ferrero, J., 2021. Do markets punish or reward corporate social responsibility decoupling? Bus. Soc. 60 (6), 1431–1467.

Gu, Y., Katz, S., Wang, X., Vasarhelyi, M., Dai, J., 2024. Government ESG reporting in smart cities. Int. J. Account. Inf. Syst. 54, 100701.

Gull, A.A., Hussain, N., Khan, S.A., Khan, Z., Saeed, A., 2023a. Governing corporate social responsibility decoupling: the effect of the governance committee on corporate social responsibility decoupling. J. Bus. Ethics 185 (2), 349–374.

Gull, A.A., Hussain, N., Khan, S.A., Mushtaq, R., Orij, R., 2023b. The power of the CEO and environmental decoupling. Bus. Strateg. Environ. 32 (6), 3951–3964.

Han, H., Shiwakoti, R.K., Jarvis, R., Mordi, C., Botchie, D., 2023. Accounting and auditing with blockchain technology and artificial Intelligence: a literature review. Int. J. Account. Inf. Syst. 48, 100598.

Helfat, C.E., Kaul, A., Ketchen Jr, D.J., Barney, J.B., Chatain, O., Singh, H., 2023. Renewing the resource-based view: new contexts, new concepts, and new methods. Strateg. Manag. J. 44 (6), 1357–1390.

Hine, E., Floridi, L., 2024. Artificial intelligence with American values and chinese characteristics: a comparative analysis of American and chinese governmental AI policies. AI & Soc. 39 (1), 257–278.

Hinton, T. (2021). *Understanding ESG Data and Scores from Refinitiv*. https://lsegissuerservices.com/spark-insights/understanding-refinitiv-esg-data-scores/understanding-esg-data-and-scores-from-refinitiv.

Holmes, A.F., Douglass, A., 2022. Artificial intelligence: reshaping the accounting profession and the disruption to accounting education. J. Emerging Technol. Account. 19 (1), 53–68.

iDigital. (2023). 2022 China Artificial Intelligence Industry Research Report. https://www.idigital.com.cn/report/4147?type=0.

International Monetary Fund. (2023). *AI Terminology Glossary*. https://www.imf.org/zh/publications/fandd/issues/2023/12/ai-lexicon.

Ioannou, I., Kassinis, G., Papagiannakis, G., 2023. The impact of perceived greenwashing on customer satisfaction and the contingent role of capability reputation. J. Bus. Ethics 185 (2), 333–347.

Jensen, M.C., Meckling, W.H., 1976. Theory of the firm: managerial behavior, agency costs and ownership structure. J. Financ. Econ. 3 (4), 305–360.

Jeong, A.Y., Kim, H.M., Lim, J.H., 2025. Does Blockchain help make the world better? Analyzing the effect of blockchain adoption on environmental, social, and governance performance of firms. J. Inf. Syst. 39 (2), 111–138.
Kaplan, A., Haenlein, M., 2019. Siri, Siri, in my hand: Who's the fairest in the land? on the interpretations, illustrations, and implications of artificial intelligence. Bus. Horiz. 62 (1), 15–25.
Kellogg, K.C., Valentine, M.A., Christin, A., 2020. Algorithms at work: the new contested terrain of control. Acad. Manag. Ann. 14 (1), 366–410.
Kim, E.H., Lyon, T.P., 2015. Greenwash vs. brownwash: exaggeration and undue modesty in corporate sustainability disclosure. Organ. Sci. 26 (3), 705–723.
Kolbjørnsrud, V., Amico, R., Thomas, R.J., 2016. How artificial intelligence will redefine management. Harv. Bus. Rev. 2 (1), 3–10.
Krueger, P., Sautner, Z., Tang, D.Y., Zhong, R., 2024. The effects of mandatory ESG disclosure around the world. J. Account. Res. 62 (5), 1795–1847.
Lagasio, V., 2024. ESG-washing detection in corporate sustainability reports. Int. Rev. Financ. Anal. 96, 103742.
Larcker, D.F., Rusticus, T.O., 2010. On the use of instrumental variables in accounting research. J. Account. Econ. 49 (3), 186–205.
Li, J., Wu, T., Hu, B., Pan, D., Zhou, Y., 2025a. Artificial intelligence and corporate ESG performance. Int. Rev. Financ. Anal. 102, 104036.
Li, N., Kim, M., Dai, J., Vasarhelyi, M.A., 2024a. Using artificial intelligence in ESG assurance. J. Emerging Technol. Account. 21 (2), 83–99.
Li, S., Younas, M.W., Maqsood, U.S., Zahid, R.A., 2024b. Impact of AI adoption on ESG performance: evidence from chinese firms. Energy & Environment, forthcoming.
Li, Y., Wang, Y., Romero, S., 2025b. Attention Management by visual salience in environmental, social, and governance report. J. Inf. Syst. 1–26.
Lin, B., Zhu, Y., 2025. Does AI elevate corporate ESG performance? a supply chain perspective. Bus. Strateg. Environ. 34 (1), 586–597.
Liu, X., Cifuentes-Faura, J., Zhao, S., Wang, L., 2024. The impact of government environmental attention on firms' ESG performance: evidence from China. Res. Int. Bus. Financ. 67, 102124.
Liu, Y., Song, J., Zhou, B., Liu, J., 2025. Artificial intelligence applications and corporate ESG performance. International Review of Economics & Finance 104, 104559.
London Stock Exchange Group (LSEG). (2024, October). *Environmental, Social and Governance Scores*. https://www.lseg.com/content/dam/data-analytics/en_us/documents/methodology/lseg-esg-scores-methodology.pdf.
Long, L., Wang, C., Zhang, M., 2025. Does social media pressure induce corporate hypocrisy? evidence of ESG greenwashing from China. J. Bus. Ethics 197 (2), 311–338.
Lublóy, Á., Keresztúri, J.L., Berlinger, E., 2025. Quantifying firm-level greenwashing: a systematic literature review. J. Environ. Manage. 373, 123399.
McCulloch, W.S., Pitts, W., 1943. A logical calculus of the ideas immanent in nervous activity. Bull. Math. Biophys. 5 (4), 115–133.
Mitchell, M., 2021. *Why AI is harder than we think*. arXIV, Cornell University, Ithaca, NY.
Mitchell, R.K., Agle, B.R., Wood, D.J., 1997. Toward a theory of stakeholder identification and salience: defining the principle of who and what really counts. Acad. Manag. Rev. 22 (4), 853–886.
Peng, H.R., Ling, K., Zhang, Y.J., 2024. The carbon emission reduction effect of digital infrastructure development: evidence from the broadband China policy. J. Clean. Prod. 434, 140060.
Petersen, M.A., 2009. Estimating standard errors in finance panel data sets: comparing approaches. Rev. Financ. Stud. 22 (1), 435–480.
RepRisk. (2021). *RepRisk Methodology Overview*. https://www.reprisk.com/insights/resources/methodology.
Sari, R.C., Sholihin, M., Cahaya, F.R., Ishola, A., Yuniarti, N., Pranesti, A., Sari, A.R., Haryanto, 2025. Artificial intelligence-based ESG greenwashing detection: road to net zero carbon and its impact on corporate performance. Business Strategy & Development 8 (4), e70228.
Simon, H.A., 1995. Artificial intelligence: an empirical science. Artif. Intell. 77 (1), 95–127.
Song, Z., Tsang, H.S.H., Hsung, R.T.C., Zhu, Y., Lo, W.L., 2025. From market volatility to predictive insight: an adaptive transformer–RL framework for sentiment-driven financial time-series forecasting. Forecasting 7 (4), 55.
Stratopoulos, T.C., Wang, V.X., 2025. Artificial intelligence and accounting research: a framework and agenda. Int. J. Account. Inf. Syst. 56, 100760.
Suárez Giri, F., Sánchez Chaparro, T., 2024. Unveiling the blackbox within ESG ratings' blackbox: Toward a framework for analyzing AI adoption and its impacts. Business Strategy & Development 7 (4), e70038.
Sutton, S.G., Holt, M., Arnold, V., 2016. "The reports of my death are greatly exaggerated"—Artificial intelligence research in accounting. Int. J. Account. Inf. Syst. 22, 60–73.
Tang, P., Wang, C., Jiang, Q., Liu, X., Wang, J., 2023. Symbol or substance? Environmental regulations and corporate environmental actions decoupling. J. Environ. Manage. 346, 118950.
Tashman, P., Marano, V., Kostova, T., 2019. Walking the walk or talking the talk? corporate social responsibility decoupling in emerging market multinationals. J. Int. Bus. Stud. 50, 153–171.
Tian, H., Zhao, L., Yunfang, L., Wang, W., 2023. Can enterprise green technology innovation performance achieve "corner overtaking" by using artificial intelligence?—Evidence from chinese manufacturing enterprises. Technol. Forecast. Soc. Chang. 194, 122732.
Tian, J., Sun, S., Cao, W., Bu, D., Xue, R., 2024. Make every dollar count: the impact of green credit regulation on corporate green investment efficiency. Energy Econ. 130, 107307.
Topal, İ., Nart, S., Akar, C., Erkollar, A., 2020. The effect of greenwashing on online consumer engagement: a comparative study in France, Germany, Turkey, and the United Kingdom. Bus. Strateg. Environ. 29 (2), 465–480.
Torelli, R., Balluchi, F., Lazzini, A., 2020. Greenwashing and environmental communication: effects on stakeholders' perceptions. Bus. Strateg. Environ. 29 (2), 407–421.
Townsend, D.M., Hunt, R.A., Rady, J., Manocha, P., Jin, J.H., 2025. Are the futures computable? Knightian uncertainty and artificial intelligence. Acad. Manag. Rev. 50 (2), 415–440.
Wagner, T., Lutz, R.J., Weitz, B.A., 2009. Corporate hypocrisy: overcoming the threat of inconsistent corporate social responsibility perceptions. J. Mark. 73 (6), 77–91.
Walker, K., Wan, F., 2012. The harm of symbolic actions and green-washing: corporate actions and communications on environmental performance and their financial implications. J. Bus. Ethics 109 (2), 227–242.
Wang, Q., Hu, A., Tian, Z., 2022. Digital transformation and electricity consumption: evidence from the Broadband China pilot policy. Energy Econ. 115, 106346.
Wang, Y., Hu, F., Wang, Y., 2024. Analyst coverage and greenwashing: evidence from Chinese A-Share listed corporations. International Review of Economics & Finance 94, 103423.
World Economic Forum. (2022). *The Global Risks Report 2022*. https://www3.weforum.org/docs/WEF_The_Global_Risks_Report_2022.pdf.
Xing, C., Zhang, Y., Tripe, D., 2021. Green credit policy and corporate access to bank loans in China: the role of environmental disclosure and green innovation. Int. Rev. Financ. Anal. 77, 101838.
Yang, G., Yang, X., 2025. AI adoption and ESG performance: evidence from China. International Review of Economics & Finance 104659.
Zhang, C., Cho, S., Vasarhelyi, M.A., 2022. Explainable artificial intelligence (XAI) in auditing. Int. J. Account. Inf. Syst. 46, 100572.
Zhang, C., Yang, J., 2024. Artificial intelligence and corporate ESG performance. International Review of Economics & Finance 96, 103713.
Zhang Parker, C., Jiang, L., Cho, S., Vasarhelyi, M.A., 2025. Predicting material misstatements using machine learning. Account. Rev. 100 (6), 225–262.
Zhong, K., Song, L., 2025. Artificial intelligence adoption and corporate green innovation capability. Financ. Res. Lett. 72, 106480.
Zhou, X., Li, G., Wang, Q., Li, Y., Zhou, D., 2025. Artificial intelligence, corporate information governance and ESG performance: Quasi-experimental evidence from China. Int. Rev. Financ. Anal. 102, 104087.